

Inorganic chemistry

Laboratory Manual I

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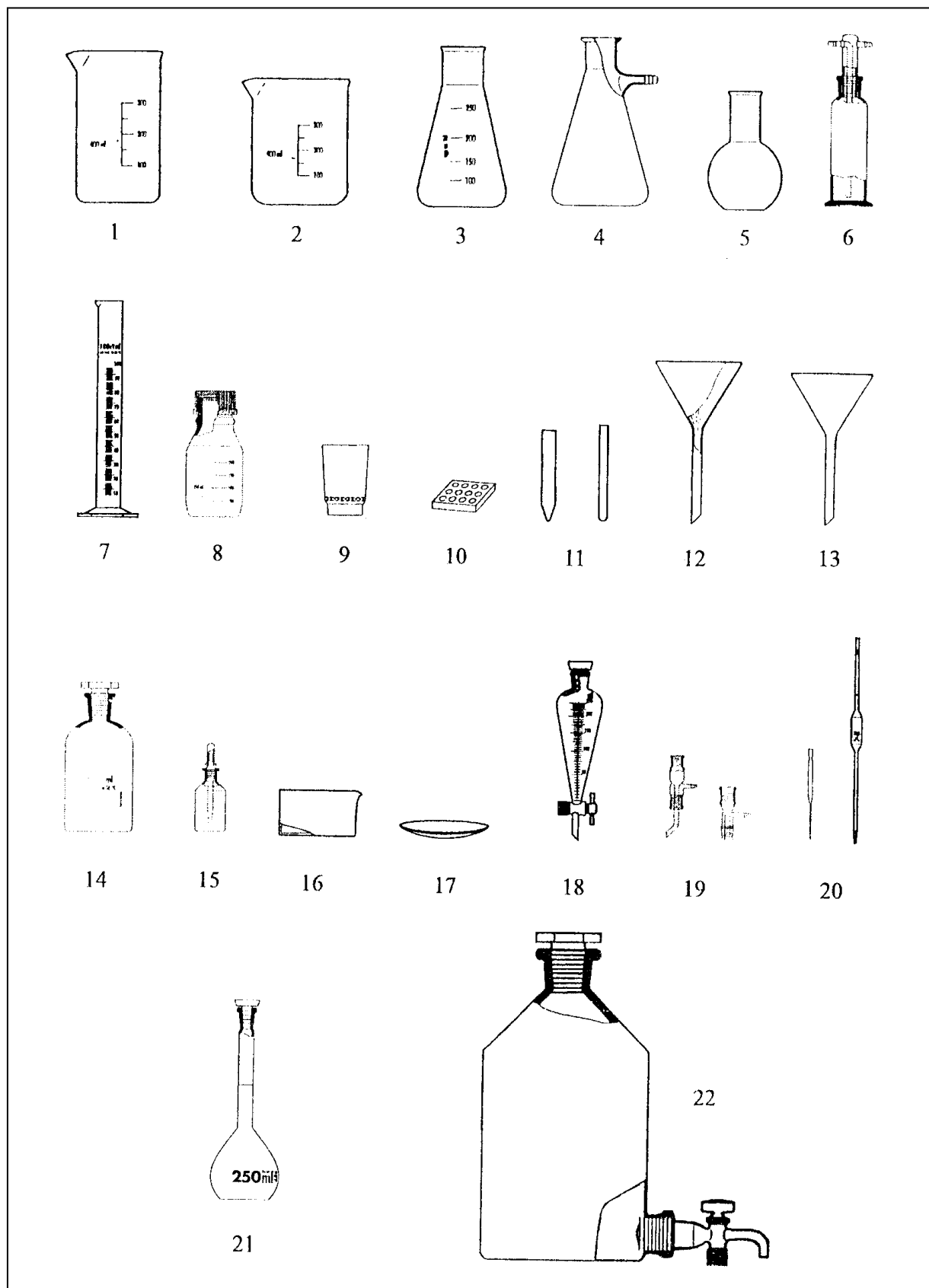
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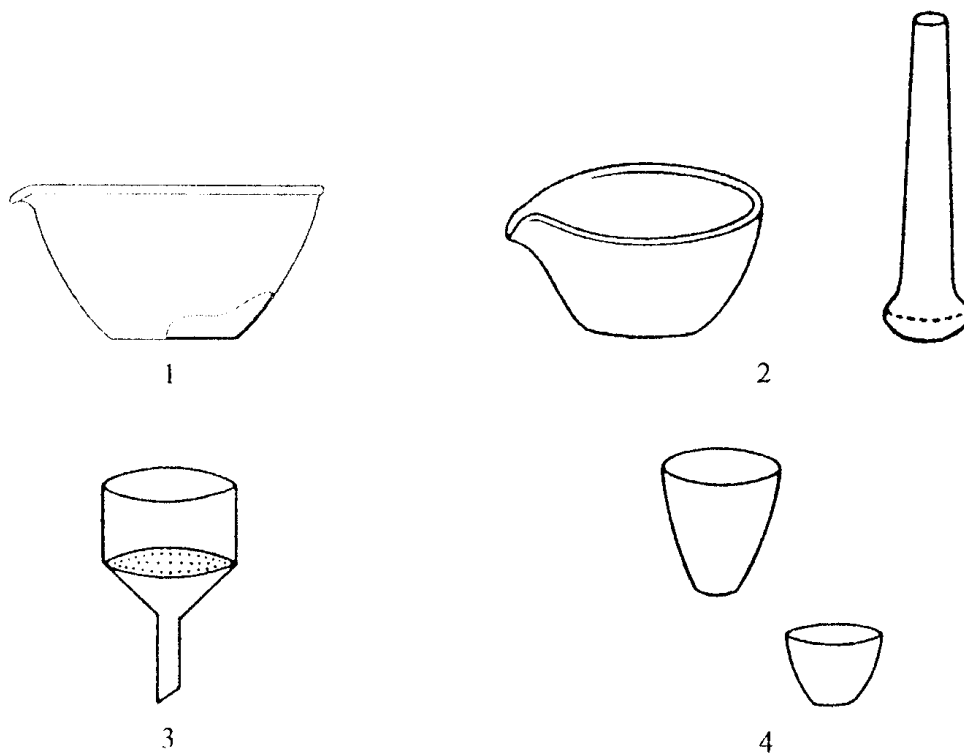
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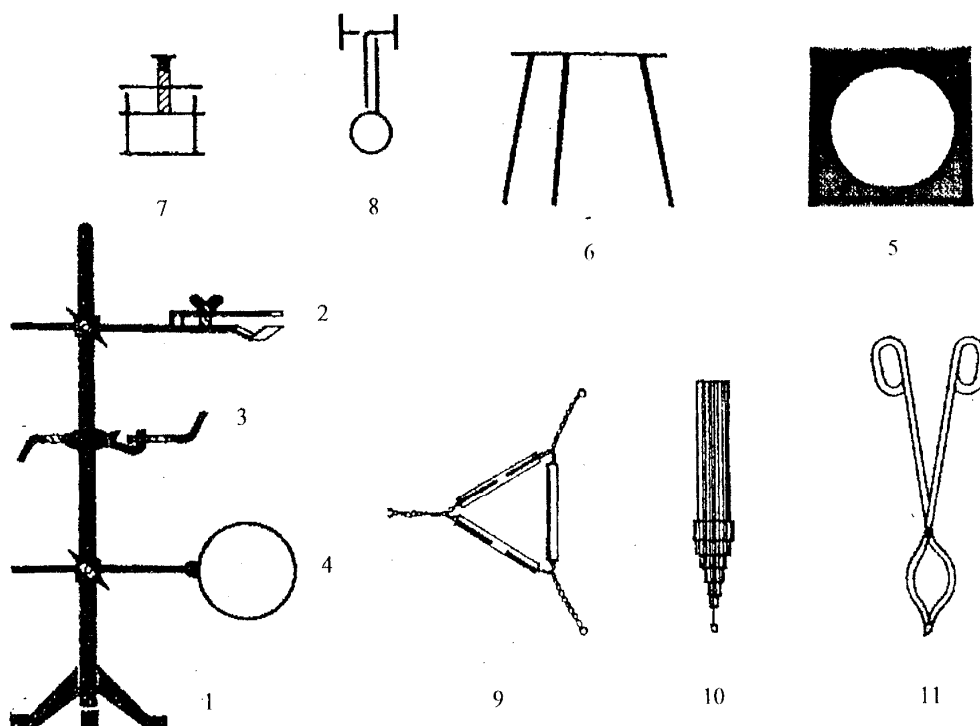


Laboratory glassware

1,2-beakers, 3-Erlenmeyer flask, 4-filter flask, 5-boiling flask, 6-gas washing bottle, 7-graduated cylinder, 8-bottle, 9-fritted glass, 10-spot plate, 11-test tube, 12,13-funnels, 14-reagent bottle, 15-dropping bottle, 16-crystallization dish, 17-watch glass, 18-separatory funnel, 19-receivers, 20-pipets, 21-volumetric flask, 22-bottle with stopcock drain



Laboratory porcelain ware: 1-evaporating dish, 2-mortar and pestle, 3-Büchner funnel, 4-crucibles



Laboratory hardware: 1-stand, 2-clamp, 3-clamp holder, 4-ring, 5-wire gauze, 6-tripod, 7-screw clamp, 8-pinchcock clamp, 9-clay triangle, 10-cork borer, 11-crucible tongs

Laboratory Rules and Safety

1. Eye protection should be worn when you are working in the laboratory.
2. Laboratory coats should be worn to protect clothing.
3. Students with long hair should tie it away from the face.
4. Follow the instructions given in the description of experiments. Never perform unauthorized experiments.
5. Know the location of fire extinguishers, fire blankets, eyewashes, showers and first aid kits.
6. Read all labels before using chemicals. Do not take more chemicals than you need. Never return any unused material to the storage bottle.
7. NEVER use flammable liquids near a flame or an ignition source.
8. When heating a liquid in a test tube or carrying out a reaction in a test tube, never point the mouth of the test tube at yourself or your neighbour.
9. Never pour water into concentrated acid. Always slowly pour the acid into water while mixing.
10. If dangerous gases are produced during the experiment, do the experiment under the hood.
11. NEVER inhale gaseous fumes. If you need to determine the odour of a gas, fan a small amount of the vapour towards your nose with your hand.
12. Return all waste chemicals to the containers indicated by your instructor. If no containers are available, pour all waste liquids directly down the drain and then run water down to drain.
13. NEVER put broken glass into undesignated wastebaskets.
14. Keep the balance area clean and free of spilled chemicals or used filter papers.
15. If you splash any acid or base in your eyes, face or hands, flush [it](#) with plenty of water.
16. Report all accidents or injuries (cuts and burns) to the laboratory instructor immediately.
17. Eating or drinking in the laboratory is strictly prohibited.
18. Wash your hands before leaving the laboratory.
19. Any student who is sensitive to chemicals is strongly advised to check with his/her doctor if this course may pose any hazard to their health.
20. Students who may be pregnant cannot work in the laboratory.

Laboratory reports

1. **The Title Page** consists the name of the experiment, name of the student and the date.
2. **The Introduction** states the objective of the experiment and provides the reader with background of the experiment. State the topic of your report clearly and concisely, in one or two sentences and give chemical reactions.
3. **Methods and Materials (or Equipment)**
4. **Results** are usually dominated by calculations, tables and figures; however, you still need to state all significant results explicitly in verbal form.
Compare expected results with those obtained.
Analyze experimental error.

Categories of danger

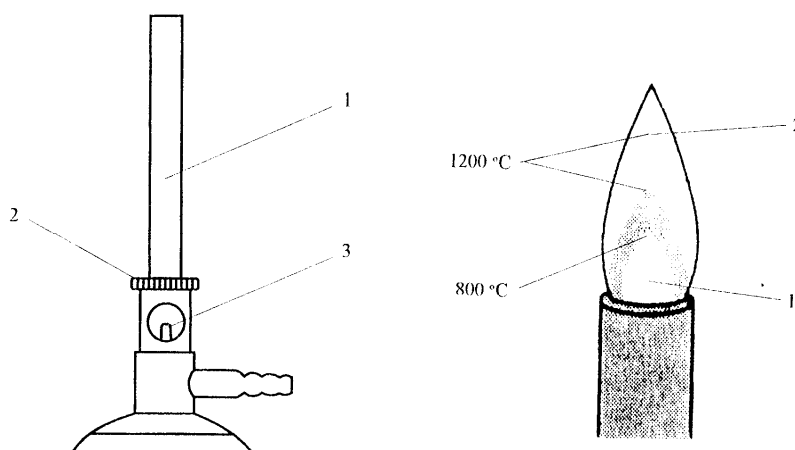
Explosive, E (Chemicals that explode)	
Oxidising, O (Chemicals that react exothermally with other chemicals)	
Extremely Flammable, F+, or Highly Flammable, F, or Flammable (Chemicals that have an extremely low flash point and boiling point, and gases that catch fire in contact with air OR chemicals that may catch fire in contact with air, only need brief contact with an ignition source, have a very low flash point or evolve highly flammable gases in contact with water.)	
Very Toxic, T+, or Toxic, T (Chemicals that in very small amounts cause damage to health OR Chemicals that in small amounts cause damage to health)	
Harmful (Chemicals that may cause damage to health)	
Corrosive, C (Chemicals that may destroy living tissue on contact)	
- Irritant, Xi - Sensitising, Xn or Xi (Chemicals that may cause inflammation to the skin or other mucous membranes)	
- Carcinogenic, Categories 1 and 2, T - Carcinogenic, Category 3, Xn (Chemicals that may cause cancer or increase its incidence)	
Dangerous for the Environment, N (Chemicals that may present an immediate or delayed danger to one or more components of the environment)	

Laboratory techniques

Heating

A *Bunsen burner* has a weighted base where the gas supply attaches and a vertical metal tube through which a fine jet of fuel gas is directed. The gas flows from the gas supply connection to the base. The stream of gas then passes through a small hole at the bottom of the tube and is directed upward through the tube. Air is drawn in through air holes near the base of the tube and the mixture is ignited and burns at the tube's upper opening.

Adjusting the Burner - after lighting the burner, it must be properly adjusted. In a properly adjusted Bunsen burner, there will be a blue flame containing two or more cones.



1-tube, 2-air supply, 3-gas jet

1-inner blue cone (reducing), 2-outer cone (oxidizing)

A Bunsen burner can produce 3 different types of flames:

The "coolest" flame is of yellow/orange colour. It is approximately 300 °C. It is never used to heat anything, only to show that the Bunsen burner is on. It is called the safety flame.	The medium flame, also called the blue flame or the invisible flame is difficult to see in a well-lit room. It is the most commonly used flame. It is approximately 500 °C.	The hottest flame is called the roaring blue flame. It is characterized by a light blue triangle in the middle and it is the only flame of the 3 which makes a noise. It is approximately 700°C.
		

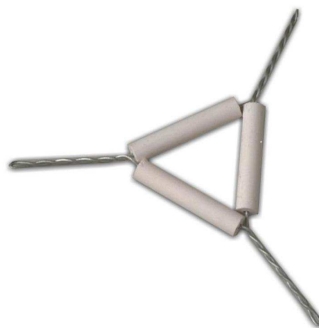
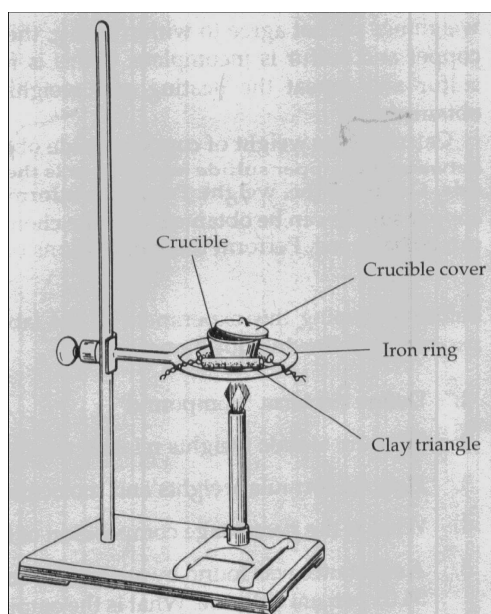
The mixture of air and gas (optimally about 1 part of gas to 3 parts of air) is forced by gas pressure to the top of the tube, where it is ignited with a match. It burns with a pale blue flame - the primary flame (seen as a small inner cone) and a secondary, almost colourless flame (seen as a larger, outer cone, which appears when the remaining gas is completely oxidized by the surrounding air).

The hottest part of the Bunsen flame, which is found just above the tip of the primary flame, reaches about 1500 °C. When too little air is drawn in, the gas mixture will not burn completely and it will form tiny carbon particles that are heated to glowing, making the flame luminous. With too much air, the flame may burn inside the burner tube; that is, it may strike inside.

Heating of solids – annealing

Techniques where crucibles are used - crucibles are usually made out of porcelain and they may be heated to very high temperatures, either by a direct Bunsen burner flame or in an oven, without breaking. The crucible should be heated slowly with the Bunsen burner held at a distance at first and then brought in closer as the crucible gets warmer. The crucible can be heated until it is red hot. Always place the hot crucible on a wire gauze, NEVER directly on the desk top. The cold surface may cause cracking of the crucible

Setup a ring stand with an iron ring and a clay triangle. The height should be adjusted so that the hottest part of the flame (the tip of the inner blue cone) is touching the crucible when the crucible is held by the triangle.



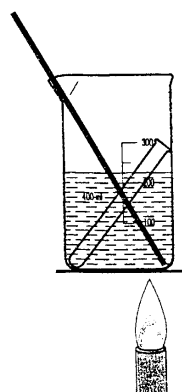
the clay triangle is used to hold a crucible while the crucible is heated.

Heating of liquids

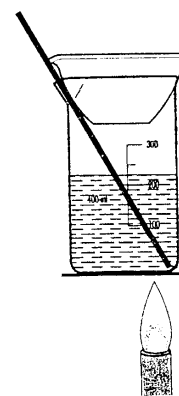
- test tube should be no more than one-third full, place a gentle flame at the same level as the top of the liquid, not at the base
- beaker or flask lay on a wire gauze, place a glass stirring rod or boiling chip in the beaker (to minimise the problem of bumping)
- a hot water bath (a)



Safe heating in a test tube



a
water bath



b
steam bath

Evaporation of liquids

- from an evaporating dish over a steam bath (b)

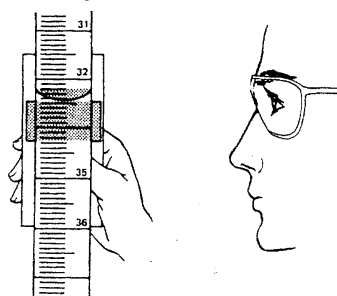
Weighing

- chemicals should never be placed directly on the balance pan
- use a beaker or watch glass to measure the mass of laboratory chemicals
- do not drop anything on the pan

Liquid volume

Equipment for measurement of volume: graduated cylinder, measuring (Pasteur) pipet, volumetric pipet, buret, volumetric flask.

The curved surface of a liquid is called the meniscus. When reading, be sure to have your eye at the same level as the meniscus in order to avoid problems due to parallax.

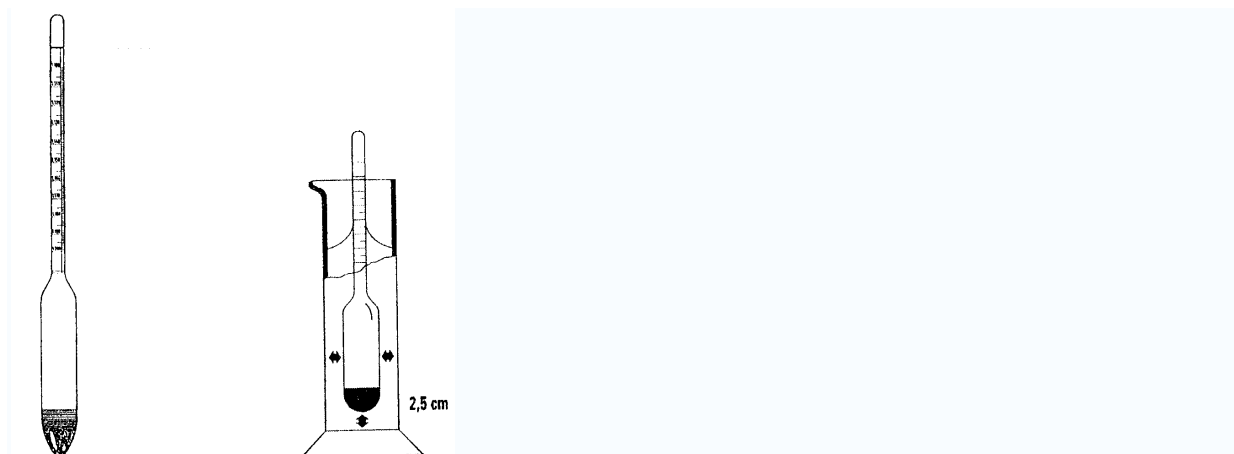


Reading a meniscus

Density (symbol: ρ - Greek: rho) is a measure of mass per unit of volume. The higher an object's density, the higher its mass per volume. The *average density* of an object equals its total mass divided by its total volume. The **SI unit** of density is kilogram per cubic meter (**kg/m³**)

$$\rho = \frac{m}{V}$$

where ρ is the object's density (measured in kilograms per cubic meter)
 m is the object's total mass (measured in kilograms)
 V is the object's total volume (measured in cubic meters)



Densimeter and measurement of density

Separation of liquid and solid phases

Decanting, or pouring of the liquid, has two major functions in the laboratory. The first is to simply measure out a quantity of liquid, usually from a reagent bottle. The second is pouring off a supernatant liquid, to separate it from a precipitated solid.

- allow the solid to settle down on the bottom of the beaker, then transfer slowly the liquid with the aid of a stirring rod (not to disturb the settled solid).

Filtration

Gravity filtration is most useful when the solid is to be discarded and the filtrate (the liquid that flows through the filter paper) is the desired product.

- fold the filter paper and place it into the funnel, steady the funnel with the support clamp, moisten the folded filter paper with the solvent being filtered, the funnel's tip should touch wall of the beaker to reduce splashing,
To prepare a filter paper for gravity filtration, fold it in half, and then in quarter (with one quarter slightly smaller than the other). Tear off a corner of the smaller quarter, and open the cone so that there are three layers on one side, and one layer on the other.

Vacuum filtration is often faster than gravity filtration and is most useful if the solid must be kept or if a filtration must be done rapidly.

- a Büchner funnel (connected with filter flask by rubber stopper) is used, a disc of filter paper fits over flat, perforated bottom of this funnel, the vacuum is created by a water aspirator. A water trap must be placed between the filter flask and the aspirator to prevent water from backing up into the filter flask.

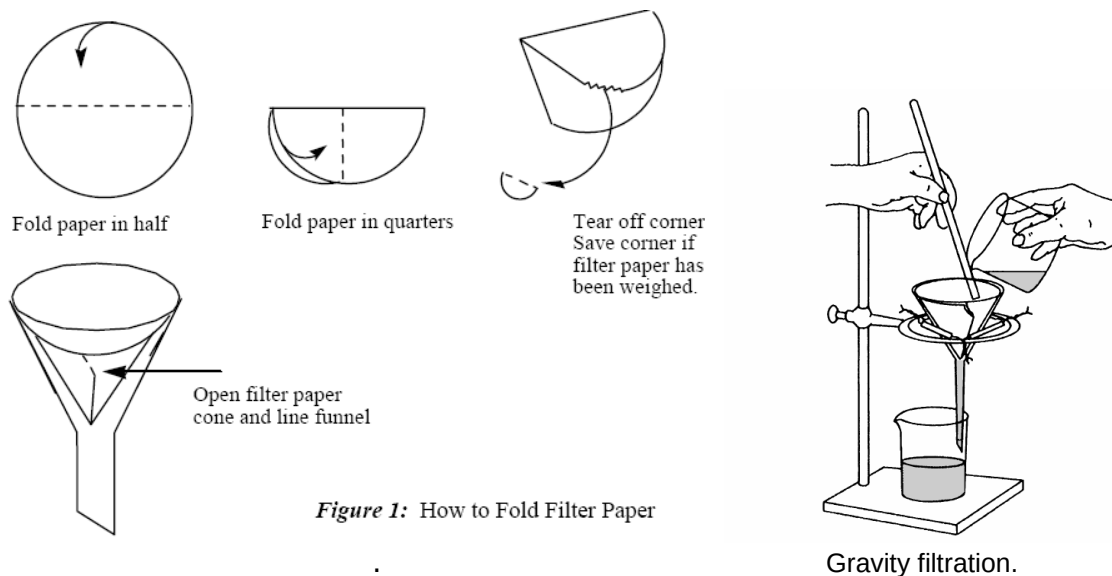
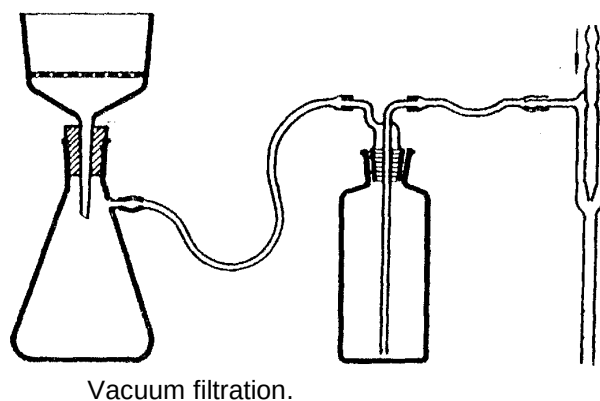
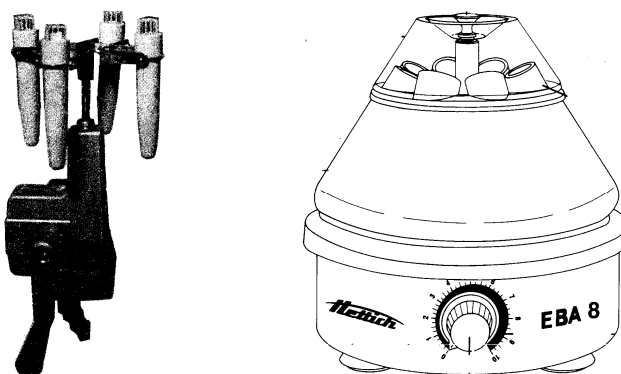


Figure 1: How to Fold Filter Paper



Centrifugation is a process that use the centrifugal force for the separation of mixtures.

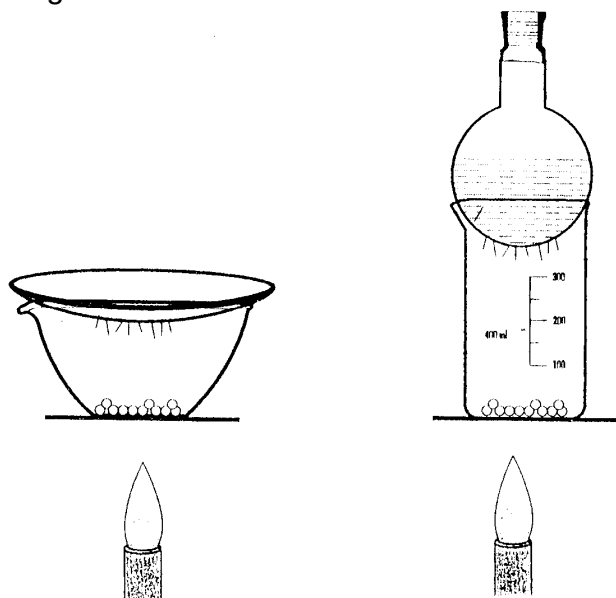
A centrifuge is a piece of equipment that puts a substance in rotation around a fixed axis in order to separate a fluid from a solid substance by centrifugal force. The load in a laboratory centrifuge must be carefully balanced.



Laboratory centrifuges

Crystallization is essentially solid-liquid separation technique. It is a process of formation of solid crystals from a homogeneous solution. This method is used to isolate and purify the required component of a reaction mixture. For crystallization to occur, the solution ought to be supersaturated. This can be achieved by solvent evaporation or cooling.

Sublimation of an element or substance is a conversion between the solid and the gaseous phases of matter, with no intermediate liquid stage. The element (iodine) is sublimed by heating of the reaction vessel and crystallization of the (iodine) vapour using an ice-cooled watch glass.



Preparation of gases

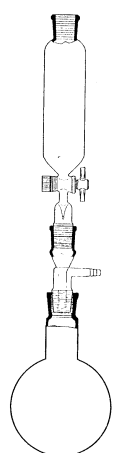
Gases are produced as products in many chemical reactions. Several common elements (H_2 , He, N_2 , O_2 , F_2 , Cl_2) and compounds (NH_3 , CO_2 , H_2S , SO_2) are present in the form of gases at room temperature.

The gases are prepared by temperature decomposition of solid compounds (NH_4NO_2 , $\text{Pb}(\text{NO}_3)_2$, HgO) or by reaction of solid compounds with liquid ($\text{CaCO}_3 + \text{HCl}$, $\text{FeS} + \text{HCl}$, $\text{Zn} + \text{HCl}$).

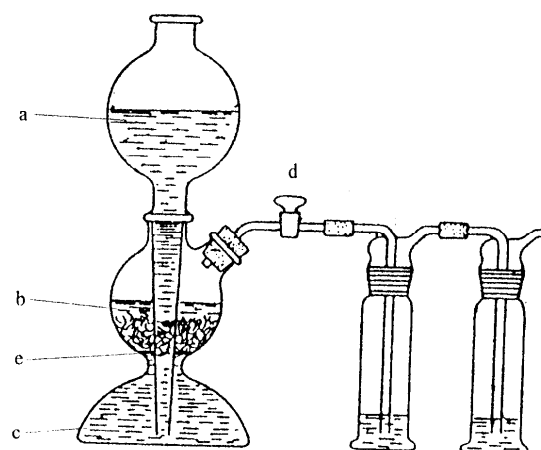
The suitable apparatus depends on the desired amount of evolved gas.

One of the most convenient apparatus for preparation of larger amounts of gas, or when it is desired to have the gas always ready for use, is the **Kipp generator**.

The apparatus is made of three vertically stacked cylinders. The solid material (eg. iron sulfide) is placed into the middle cylinder (b), the acid is put into the top cylinder (a). A tube (e) extends from the top cylinder into the bottom cylinder. The middle cylinder has a tube with a stopcock (d) attached, which is used to draw off the evolved gas. When the stopcock is closed, the pressure of the gas in the middle cylinder rises and expels the acid back into the top cylinder, until it is not in contact with the solid material anymore, and the chemical reaction stops.



gas production apparatus



Kipp apparatus

Compressed Gas Cylinders

In most laboratory applications, gases are bled from compressed gas cylinders into systems that are near atmospheric pressure and that are provided with a sort of safety outlet in case an exceedingly high pressure is reached.

There are three cases: i) the substances that remain gaseous at standard temperature and increased pressure (acetylene, air, argon, helium, oxygen), ii) the substances that are liquidized at standard temperature and increased pressure (butane, propane) and iii) the substances that are liquidized at reduced temperature and increased pressure (liquid He, N₂, O₂).

Cylinder shoulder colours

By hazard property

Flammable	Red	
Toxic/corrosive	Yellow	
Inert	Bright green	
Oxidising	Pale blue	

Note: More than one hazard property may be shown on the cylinder shoulder e.g. red and yellow

By specific gas

Argon	Dark green	
Carbon dioxide	Dusty grey	
Helium	Brown	
Nitrogen	Black	
Nitrous oxide	Dark blue	
Oxygen	White	



PROPER PROCEDURE OF OPENING OF CYLINDER VALVES

Make sure that the regulator outlet valve (**A**) is shut. Screw it clockwise until it seats. Do not over tighten it or you may damage the valve seat.

Make sure that the regulator control valve (**B**) is shut.

Open the tank valve slowly (counter clockwise). Watch the tank pressure on the regulator (**C**).

Slowly turn the regulator control valve (**B**) until the regulator pressure (**D**) is at the desired level.

Open the regulator outlet valve (**A**). You can regulate flow with this valve, but the ultimate pressure depends on the setting of the regulator control valve!

Solutions and solubility

The *solubility* of a solution is a measure of how much of the solid can be dissolved into the solvent. The solution reaches a point called the *saturation point* when no more solid will be accepted by the solvent. Any further addition of solid will result in appearance of dispersed solid mixed in the saturated solution. Each solvent and solute pair has a characteristic solubility at a given temperature. Usually as you increase the temperature, an increased amount of solid will be able to dissolve.

A *solution* is a homogeneous, liquid mixture of two or more substances.

A *solvent* is a dissolving agent, e.g. water.

A *solute* is a substance that is dissolved in a solution.

Water is a polar molecule. It will dissolve ionic compounds. It will also dissolve some molecular compounds that have covalent polar bonds.

Strong and Weak Electrolytes

Electrolytes are substances whose aqueous solutions conduct electricity. It is the presence of ions, which can move through the solution that allows conduction of the electricity. Sometimes the ions were already present in the ionic salt, or they result from the reaction of a substance with a solvent, water.

Strong electrolytes are substances which are completely dissociated in water or substances for which the ionization reaction goes to completion (soluble salts, strong inorganic acids, strong inorganic bases)

Weak electrolytes only ionize partially in water (weak inorganic acids, organic acids, ammonia).

CLASSES OF SOLUBLE COMPOUNDS	CLASSES OF INSOLUBLE COMPOUNDS
compounds of Group I elements	carbonates (CO_3^{2-}), phosphates (PO_4^{3-}), oxalates ($\text{C}_2\text{O}_4^{2-}$), and chromates (CrO_4^{2-}) of all cations EXCEPT Group I elements and NH_4^+
ammonium (NH_4^+) compounds	sulfides (S^{2-}) of all cations EXCEPT Group I, II, and NH_4^+
chlorides (Cl^-), bromides (Br^-), and iodides (I^-) of all cations EXCEPT Ag^+ , Hg_2^{2+} , and Pb^{2+}	hydroxides (OH^-) and oxides (O^{2-}) of all cations EXCEPT Group I and II and NH_4^+
nitrates (NO_3^-), chlorates (ClO_3^-), perchlorates (ClO_4^-), and acetates (CH_3CO_2^-)	
sulfates (SO_4^{2-}) of all cations EXCEPT Sr^{2+} , Ba^{2+} , and Pb^{2+}	$\text{Ca}(\text{OH})_2$ and $\text{Sr}(\text{OH})_2$ are slightly soluble. $\text{Mg}(\text{OH})_2$ is very slightly soluble.

Colloids, sols, emulsions, gels, aerosols

(a) **Colloids** are glue-like amorphous substances, e.g. gelatine or starch, with particles bigger than most molecules, i.e. from 10^{-7} to 10^{-5} m. The particles are too large to pass through a membrane but too small to be observed under a microscope. Colloids have a dispersed phase of particles scattered through a continuous phase, the medium. Colloids are a type of mixture with properties between that of true solutions and suspensions. The colloidal particles are still too small to settle out of solution due to gravity (do not separate on standing).

(b) **Sols** are dispersions of small groups of molecules in a medium. Sols remain dispersed as the Brownian movement prevents the groups of molecules from precipitating under the influence of gravity. Lyophilic ("water loving") sols, e.g. starch in water have large dispersed particles with an affinity to the medium. Lyophobic sols, e.g. silver chloride in water, have a dispersed phase with no affinity to the medium. The particles keep apart because of their electrical charge, but eventually they precipitate.

(c) **Emulsions** are colloids where both phases (the continuous one and the dispersed one) are liquids, e.g. oil in water. When we shake two immiscible liquids, droplets of one liquid are dispersed in the other. In temporary emulsions, e.g. kerosene in water or oil in water, there is no attraction between the two liquids. The two phases will disperse in each other if shaken together but will separate on standing.

(d) **Gels** are colloids in which liquids are dispersed in solids to form a jelly. Some gels can lose the liquid component by heating to form a solid, e.g. silica gel.

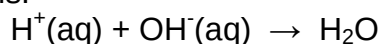
(e) **Aerosols** (fogs) are dispersions of liquids or solids in a gas, e.g. fly spray, mist, smoke.

(f) **Foams** are dispersions of gases in liquids, e.g. fire extinguisher foam or gases in solids, e.g. foam rubber mattress.

Types of chemical reactions - the main types of reactions are precipitation reactions, acid/base reactions, and redox reactions.

Precipitation reaction is a reaction in which soluble ions in separate solutions are mixed together to form an insoluble compound that settles out of solution as a solid. That insoluble compound is called a precipitate.

Acid-Base reaction is a reaction between an acid and a base which produces a neutral (pH = 7) solution (neutralization). The reaction is between the hydrogen ions and the hydroxide ions:



pH adjustment - it is often necessary to adjust the pH of a solution until it is just alkaline or just acidic. This is usually accomplished by a dropwise addition of a strong acid or base. In order to make sure that the solution does not become too acidic or basic, the pH of the solution must be monitored. You will use universal indicator paper to determine the pH of your solutions. The proper method of determination of the pH of a solution involves stirring the solution with a clean glass rod and then touching a piece of indicator paper with a tip of the rod. Do **NOT** place the indicator paper directly in your solution! You should check the pH of the solution after each addition of a drop of acid or base. If you are to acidify, stop the addition as soon as the indicator paper registers a pH of just less than 7. If the solution is to be made basic, add base until the indicator paper registers just more than 7. If you are to neutralize the solution, add the appropriate acid or base until the paper indicates the pH very close to 7.

Oxidation-Reduction Reaction - in an oxidation-reduction or redox reaction, there occurs an exchange of electrons between two reactants. The substance which loses electrons is said to be oxidized. The substance which gains those electrons is said to be reduced.

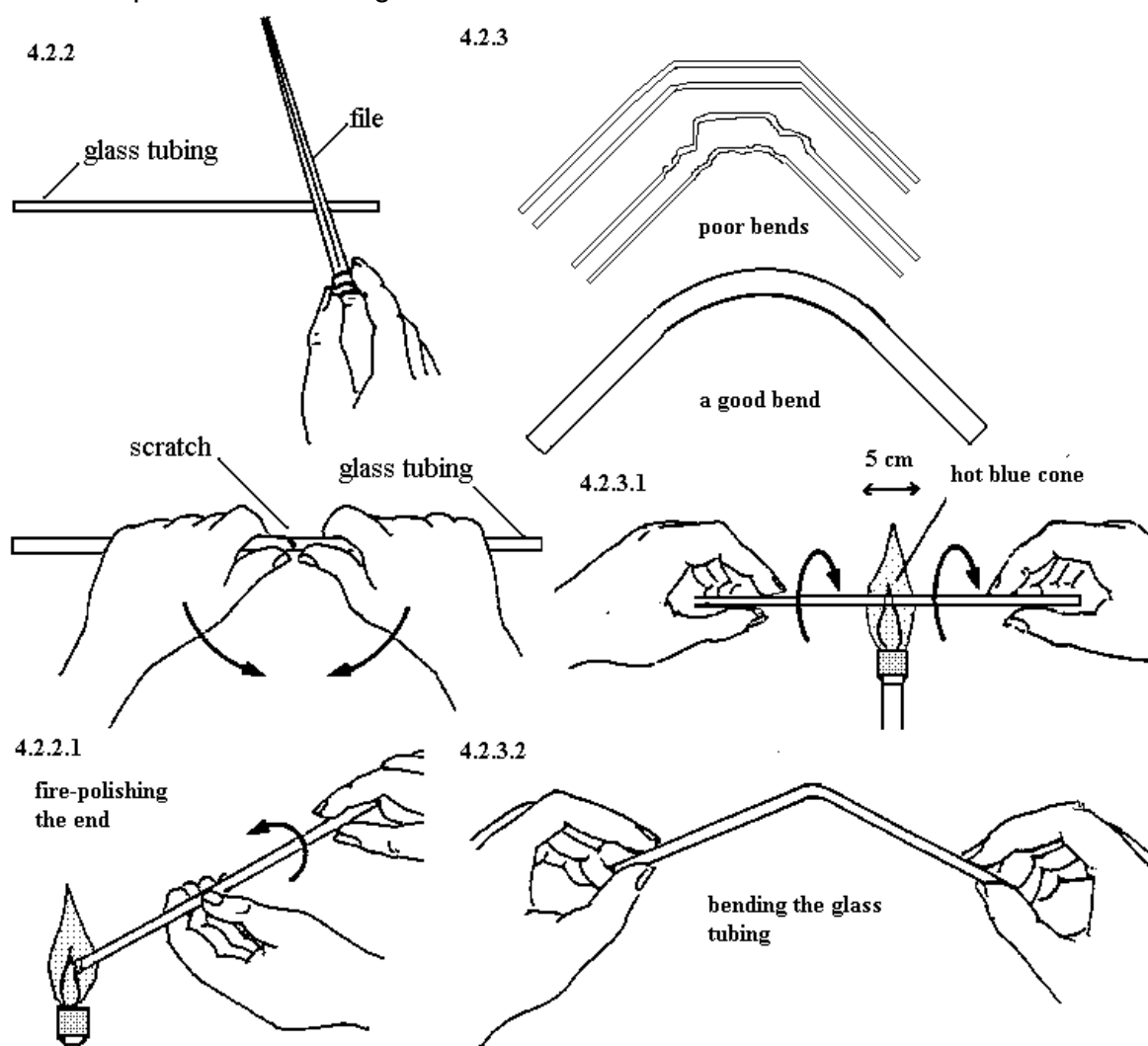
What is ... ?	Definitions
Solution	Homogeneous mixture
Solute	A substance that dissolves
Solvent	A substance in which the solute dissolves
Homogeneous mixture	A solution with one visible phase
Heterogeneous mixture	A mixture with two or more visible phases
Concentrated solution	A solution containing high percentage of the solute
Dilute solution	A solution containing low percentage of the solute
Saturated solution	A solution in which no more solute can be dissolved at a fixed temperature
Unsaturated solution	A solution in which more solute can be dissolved at a fixed temperature
Supersaturated solution	A solution containing more dissolved solute than is normal at a particular temperature
Solubility	Maximum amount of solute which will dissolve in a fixed quantity of a given solvent at a specific temperature
To dissolve	To disperse in a liquid to form a homogeneous solution
Soluble	Ability of a substance to be dissolved in a particular solvent
Insoluble	Inability of a substance to be dissolved in a particular solvent

Preparation part (number of experiment is the same as in Laboratory program)

1.1 Work with glass

Cutting of glass tubing (4.2.2). Place the tube on the table. Measure the required length. Hold the tube firmly and draw a triangular file across it a couple of times so that it makes a scratch. Do not saw back and forth. One or two firm scratches are usually sufficient. Take the tube in both hands, one each side of the scratch. Keep your thumbs as close as possible to the scratch. Press gently with your thumbs and pull with the fingers. The two pieces of tubing should separate. Do not use brute strength! If the tube does not break easily, make the file scratch a little deeper and longer. Fire-polish the cut ends (4.2.2.1).

Bending of glass tubing (4.2.3). Light the burner and open the air hole. This gives a hot blue flame. Using both hands, move the tube back and forth through the top of the flame. At the same time rotate the tube. Heat about 5 centimetres of the tube. When the tubing is warm, lower it into the dark blue cone of the flame. Keep rotating the tube until it glows red and the tube becomes soft. Take the tube out of the flame and bend it. Do not force the bending and once bent do not reheat it. Hold the tubing until it cools a little. Then put it on the wire gauze to cool.



1.2.2 Crystallization of K₂Cr₂O₇ (crystallization in motion)

Material: 30 g K₂Cr₂O₇

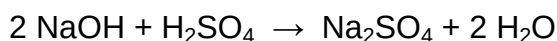
Procedure: Using the values of solubility of K₂Cr₂O₇ calculate the amount of water needed for preparation of saturated solution at 60 °C and the theoretical yield of crystals.

Solubility of K₂Cr₂O₇ at 20 °C is 12 g and at 60 °C is 43 g in 100 ml H₂O.

Measure the calculated volume of water and heat it to 80 – 90 °C in a beaker, dissolve 30 g of K₂Cr₂O₇. Hold the temperature during the dissolving near the 80 °C. Filter the solution using a Büchner funnel. Cool the filtrate to 20 °C and filter the crystals (Büchner funnel). Dry the crystals at 100 °C and weigh it, calculate the real yield (%).

Weigh and measure volume of filtrate (mother liquor) and save it for procedure 2.3.4.

1.4.1 Preparation of Na₂SO₄ · 10 H₂O (15 g)



Material: NaOH, H₂SO₄, pH indicator paper

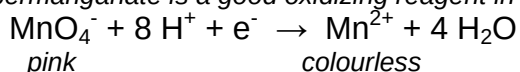
Procedure: Calculate the amount of NaOH and H₂SO₄ for preparation of 15 g Na₂SO₄ · 10 H₂O and measure/weigh them. Prepare 10 % solutions of both chemicals. The neutralization is realized by mixing about 95 % of volume of both solutions with continuous stirring. Measure pH of the mixture and adjust it to value of 6 – 8 if necessary (add drop of the appropriate solution).

Concentrate the solution of sodium sulphate through evaporation over a steam bath to crystallization. Filter the crystals and dry them between sheets of filter paper.

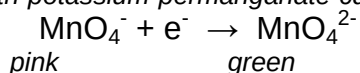
1.5.1 Reactions of usual laboratory oxidizing agents – KMnO₄ and K₂Cr₂O₇

(record your observations and write the balanced equations for all reactions)

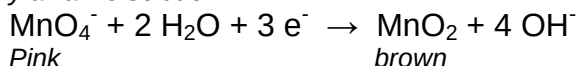
Potassium permanganate is a good oxidizing reagent in acid solution and the half-reaction is:



Oxidation with potassium permanganate can be carried out in alkaline solution. In strong alkali the half reaction is:



And in weakly alkaline solution:



Procedure:

a) Prepare solutions of oxidation agents – KMnO₄ and K₂Cr₂O₇ (the characteristic purple-violet colour of the permanganate ion is observed in 0.02 M solution of KMnO₄).

b) Pour about 1 – 2 ml of a 5 % Na₂SO₃ solution in three separate test tubes and arrange pH:

pH ≤ 7, by a drop of diluted H₂SO₄ (1:5)

pH 6 – 8

pH ≥ 8, by a drop of 10 % solution of NaOH.

Add KMnO₄ solution (dropwise) to each solution in the test tube, observe the colour changes and record the reaction.

c) Repeat this procedure with solution of K₂Cr₂O₇ as an oxidation agent.

d) Reaction b) and c) repeat with 5 % solution of NaNO₂ in place of Na₂SO₃.

2.1.4 Reactions of hydrogen peroxide (in test tubes)

(record your observations and write the balanced equations for all reactions)

a) Dissolve one crystal of KMnO_4 in water. Acidify this solution with diluted H_2SO_4 (3 drops) and add 3 % solution of H_2O_2 . Observe the colour change and test the released gas with a glowing splinter.

b) Add a solution of NH_3 to 1 ml of 5 % AgNO_3 solution until the created precipitate is dissolved. Heat the solution and add 10 % H_2O_2 (0.5 ml). Observe the film on the wall of a test tube.

c) Add dropwise 10 % solution of H_2O_2 to 3 ml of 10 % KI solution acidified with 1 ml of diluted H_2SO_4 (1:2). Observe the colour change.

d) 2 ml of 5 % $\text{Cr}_2(\text{SO}_4)_3$ solution alkalize with solution of KOH (20 %) until the created precipitate is dissolved. Add 10 % H_2O_2 solution (dropwise) and observe the colour of the solution.

e) Add solid MnO_2 into 3 ml of 10 % H_2O_2 solution and test the released gas with a glowing splinter.

f) 3 ml of 0.1 % $\text{K}_2\text{Cr}_2\text{O}_7$ solution acidify with diluted H_2SO_4 (1:2). Cool the test tube (dry ice) and add 3 % H_2O_2 solution (dropwise). Observe the colour of the solution.

2.2.1 The halogens – perform this experiment in a fume hood

(record your observations and write the balanced equations for all reactions)

a) Put about 0.1 g of KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ and CaOCl_2 into three separate test tubes and add 1–2 ml of concentrated HCl . Hold a wet starch iodide paper in the upper part of the test tubes.

b) Set up the gas production apparatus. Put 5 g of MnO_2 into the flask. Add concentrated HCl through the dropping funnel and heat the flask gently. The gas should begin to evolve and is fed into two test tubes with KBr and KI solutions. Extract the products into 2 ml of CCl_4 or CHCl_3 , shake and then allow the organic layer to separate. Observe the colour of the solutions.

c) Add 1 ml of bromine water to the diluted KI solution in the test tube, note the colour and extract the product into CCl_4 or CHCl_3 .

d) Observe solubility of small amount of I_2 (0.1 g) in approx. 2 ml of 10 % KI solution, ethanol, benzene, chloroform and acetone.

e) Mix 2 g of I_2 and 0.2 g of KI in a dry beaker. Place the beaker on a tripod, cover it with a round flask with cold water and warm gently. How is this process called?

f) Dissolve a crystal of iodine in 1 ml of 5 % NaOH solution. Acidify it with diluted H_2SO_4 (1:4) and observe the colour of solution.

2.2.2 Reactions of halides

a) Put 0.2 g of solid KF, KCl, KBr and KI into separate test tubes and add 1 ml of concentrated H_2SO_4 into each one. Test the released gas by a moist indicator paper.

b) Acidify a solution of KI by a few drops of acetic acid (1:3) and add diluted NaNO_2 solution. Observe the colour of solution. Repeat it, using KCl and KBr instead of KI.

c) Put approx. 1 ml of 0.1 M solutions of KF, KCl, KBr and KI into four separate test tubes. Add 3 - 5 drops of 1 % AgNO_3 solution. Leave the test tubes in the light for a few minutes. Observe if precipitates occur and their colour.

Detect solubility of precipitates in:

- HNO_3 (1:5)
- conc. $\text{NH}_3(\text{aq})$
- 20 % $\text{Na}_2\text{S}_2\text{O}_3$ solution
- conc. HCl

d) Add 2 ml of acetic acid (1:3) and about 0.3 g of granular zinc to the precipitate of silver halide. Stir the mixture and put the test tube in the boiling water bath for 10 minutes.

e) Add diluted H_2SO_4 (1:2) into test tube with solution of 0.1 g KI and 0.1 g KIO_3 in 3 ml of water. Observe the colour of solution.

2.2.3 Reactions of pseudo halides – SCN^- (1 mol/l KSCN solution)

a) Make the reactions 2.2.2 c) and d) with solution of KSCN.

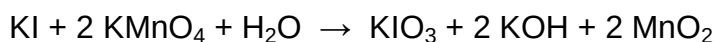
b) Add 0.5 ml of 1 M BaCl_2 solution and 0.5 ml of HNO_3 (1:2) to the 1 ml of KSCN solution. Stir the mixture and put the test tube in the water bath for a minute.

c) Add 0.5 ml of 1 M BaCl_2 solution and 0.5 ml of 6 mol/l HCl , and 1 ml 3 % H_2O_2 to 1 ml of KSCN solution. Stir the mixture and put the test tube in the water bath for a minute.

d) Add about 0.3 g of granular zinc and 0.5 ml of H_2SO_4 (1:5) to 1 ml of KSCN solution.

e) Add 2 drops of 0.1 M $\text{Fe}(\text{NO}_3)_3$ to 1 ml of KSCN solution. Observe the change of colour. Add KF solution to the solution. Describe the change and reaction.

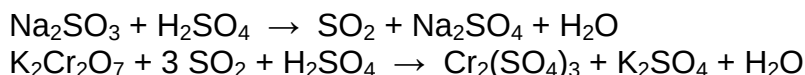
2.2.4 Preparation of potassium iodate (10 g)



Material: KMnO_4 , KI, ethanol, acetic acid

Procedure: Weight calculated amount of KMnO_4 (use 10 % excess) and dissolve it in 400 ml of warm water (use boiling flask). Add needful amount of KI in the form of saturated solution. Heat the mixture for 20 – 30 min on the water bath (colour of the solution has to be violet all the time). Add ethanol to remove excess of KMnO_4 (discolouration of solution). Filter out MnO_2 (vacuum filtration), acidify the filtrate with acetic acid and concentrate it on a steam bath to the crystallization. Filter out the KIO_3 crystals and wash them with ethanol. Wash MnO_2 properly and dry it.

2.3.4 Preparation of sulphur dioxide and its reduction effect - perform this experiment in a fume hood



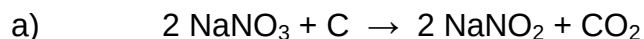
Material: Na_2SO_3 , H_2SO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ (mother liquor from task 1.2.2), NaOH

Procedure: Set up the gas production apparatus, put calculated amount of Na_2SO_3 (30 % excess) to the flask and add H_2SO_4 (1:1) through the dropping funnel. Blow the evolved gas through the gas washing bottle with 5 % $\text{K}_2\text{Cr}_2\text{O}_7$ solution (diluted mother liquor from task 1.2.2) acidified with H_2SO_4 . Bubble the unreacted gas through washing bottle with 10 % NaOH solution.

Test for end of the reaction – to the sample of reduced solution in a test tube add small amount of Na_2CO_3 crystals and heat the mixture just below the boiling point. The solution over the precipitate has to be colourless.

Add NaOH solution to the solution in the gas washing bottle to set pH to 7-8. Filter the precipitate using a Büchner funnel, wash it properly with water, dry it and weight. Calculate the practical yield and compare it with theoretical yield.

2.4.6 Preparation of sodium nitrite (10 g)



Material: NaNO_3 , C (charcoal)

Procedure: Mix calculated amounts of NaNO_3 and charcoal (10 % excess) in a dry, clean iron plate. Heat this mixture over the melting temperature of NaNO_3 . Do not add charcoal into the molten NaNO_3 ! After 10 minutes of heating cool the molten mixture on air and macerate it with hot water. Filter the unreacted charcoal and evaporate the filtrate on a steam bath to the crystallisation. Collect the crystals by filtration on the Büchner funnel and use the filtrate for task 2.4.8.



Material: NaNO_3 , Pb, $\text{CO}_2(\text{g})$

Procedure: Melt calculated amount of NaNO_3 in a dry, clean iron plate. Stir the melt with an iron rod and add Pb (5 % excess) in parts. Finish the heating and stir until solidification. Macerate the mixture with hot water, filter PbO out, wash and dry it. Blow $\text{CO}_2(\text{g})$ through hot filtrate (5 min) to remove Pb^{2+} in form of PbCO_3 and filter it out. Evaporate the filtrate on a steam bath to the crystallisation. Collect the crystals by filtration on a Büchner funnel and use the filtrate for task 2.4.8.

4.2.9 Preparation of potassium chromate

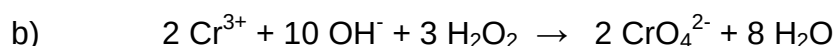


Material: KNO_3 , KOH , Cr_2O_3 (product from 2.3.4)

Procedure: Use KNO_3 and KOH in 10 % excess to the stoichiometry. Melt KOH in a dry, clean iron plate, shut down the burner and add Cr_2O_3 . Stir the mixture with an iron rod and add KNO_3 .

Heat and stir it for about 30 minutes (observe change of the colour). The reaction time may vary from the indicated value, depending on the purity and grain size of the reactants. The reaction mixture will gradually melt down to a uniform red mass. It should turn yellow upon cooling. If it has a greenish tinge it still contains unreacted chromium oxide and the heating should be continued. When all the oxide has reacted, allow the mixture to cool below 100 °C and dissolve the solid mass in 150 ml of hot water. Filter out any unreacted chromium oxide and rust particles. Evaporate the filtrate on a steam bath to the crystallisation. Collect the crystals (nearly pure K_2CrO_4) by filtration on a Büchner funnel and dry them. Evaporate the mother liquor to get a dry mixture of products ($K_2CrO_4 + KNO_2$).

Dissolve small amount of both parts of crystals in two test tubes and add dilute H_2SO_4 . Observe change of the colour and write the balanced equations for both reactions.



Material: $Cr_2(SO_4)_3$, KOH, H_2O_2

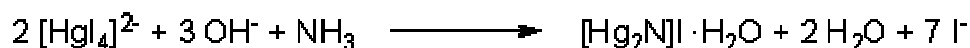
Procedure: Add 10 % KOH solution dropwise to the solution of $Cr_2(SO_4)_3$ until the $Cr(OH)_3$ precipitates. Filter it out on a Büchner funnel and wash it with a small amount of water. Put the precipitate into 600 ml beaker and add 10% KOH solution. Oxidize this suspension dropwise by 30 % H_2O_2 solution and stir. The green colour turns yellow. Boil this solution to remove excess of H_2O_2 and evaporate it on a steam bath to the crystallisation. Collect the crystals by filtration on a Büchner funnel and dry them. Prepare $Cr_2O_3 \cdot xH_2O$ (2.3.4) from the mother liquor.

2.4.7 Test for ammonia

a) Put a few NaOH granules in a mortar. Add some solid ammonium chloride and grind it up with the NaOH using a pestle. Wet litmus paper held over the mortar turns deep blue, equivalent to a pH 10-11.

b) Add a few drops of 10 % NaOH solution and 'Nessler's' reagent (5 drops) to the solution of ammonium salt.

The reaction with Nessler's reagent, $K_2[HgI_4]$, is a supersensitive test for ammonia. An alkaline solution of this compound reacts with ammonia to form a brown precipitate with the composition $[Hg_2N]I \cdot H_2O$, the iodide of Millon's base.



orange-brown

2.4.8. Reactions of nitrite and nitrate ions

(record your observations and write the balanced equations for all reactions)

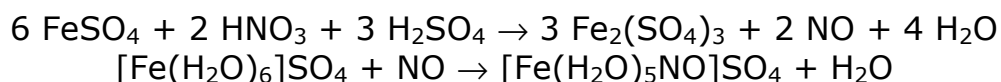
a) Add 2 – 3 drops of 10 % H_2SO_4 to the 0.5 ml of saturated $NaNO_2$ solution (mother liquor from 2.4.6). Observe blue colour of N_2O_3 , which is not stable and decomposes to NO and NO_2 .

b) Prepare 5 % $NaNO_2$ solution for reactions:

1. Add diluted H_2SO_4 (5 drops) and KMnO_4 solution (1 drop) to 1 ml of NaNO_2 solution. Repeat this reaction without H_2SO_4 .
2. Repeat reaction (1), using $\text{K}_2\text{Cr}_2\text{O}_7$ solution in place of KMnO_4 .
3. Add acetic acid (5 drops) and 5 % KI solution (5 drops) to the 1 ml of NaNO_2 solution.
4. Repeat reaction (3), using KBr solution in place of KI .
5. Add NaNO_2 solution acidified with H_2SO_4 (1:5) dropwise to 1 ml bromine water.
6. Add a small amount of $\text{NH}_2\text{SO}_3\text{H}$ crystals, acetic acid (3 drops) and 1 drop of KI solution to the 2 – 3 ml of NaNO_2 solution.
7. Heat gently a mixture of 2 ml of NaNO_2 solution, 10 % NaOH solution (5 drops) and Zn . Test the released gas with moist pH indicator paper or Nessler's reagent.
8. Add 5 drops of acetic acid and a small piece of Zn to 2 ml of NaNO_2 solution. After a while add 5 drops of KI solution.
9. Repeat reactions 1–8, using 5 % NaNO_3 solution instead of NaNO_2 .

2.4.9 Brown Ring Test

The best known qualitative test for nitrate is the "Brown Ring Test": Nitric acid is produced at the interface between the nitrate solution and the concentrated sulphuric acid. The nitric acid is reduced to nitrogen monoxide by the iron(II) ions and the brownish violet nitroso complex is then formed by nitrogen monoxide with the excess iron(II) ions.



- a) Add saturated solution of FeSO_4 to the test tube with NaNO_3 solution. Pour concentrated sulphuric acid (96 %) carefully down the inside wall of the tube. The acid sinks below the aqueous solution due to large difference in density ($\rho(\text{H}_2\text{SO}_4)=1,83 \text{ g}\cdot\text{cm}^{-3}$). After a short period an intensive brownish violet ring is formed at the phase transition.
- b) Repeat (a), using NaNO_2 solution and compare both reactions.
- c) Repeat reaction with NaNO_3 and NaNO_2 using acetic acid instead of H_2SO_4 .

2.5.2 Reactions of hypophosphite ion (use 5 % NaH_2PO_2 solution)

(record your observations and write the balanced equations for all reactions)

- a) Add 1 drop of I_2 in KI solution to 0.5 ml of NaH_2PO_2 solution.
- b) Add 1 drop of 1 % KMnO_4 solution to 0.5 ml of NaH_2PO_2 solution.
- c) Add H_2SO_4 to 0.5 ml of NaH_2PO_2 solution to set pH~3 and 1 drop of 1 % KMnO_4 solution.
- d) Add HNO_3 to 0.5 ml of NaH_2PO_2 solution to set pH~3 and 1 drop of 5 % AgNO_3 solution.

2.5.1 Preparation of phosphine (demonstration) - perform this experiment in a fume hood



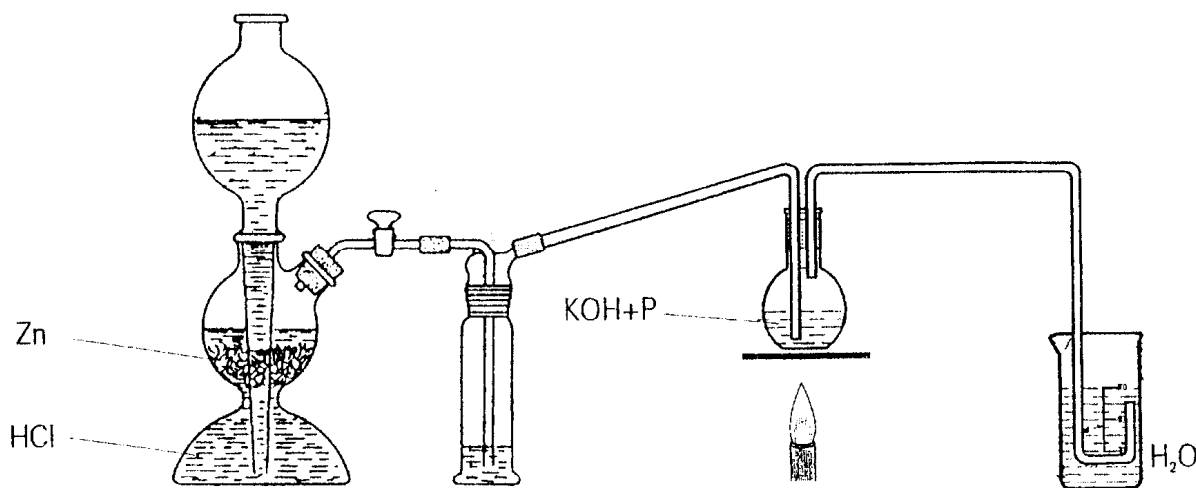
Material: P_4 , KOH, Zn, HCl

Procedure: Arrange an apparatus as represented in the figure. Fill a Kipp generator with sticks of Zn and put HCl (1:1) into the acid reservoir.

Place 100 ml of 5 % KOH solution and about 1 g of white phosphorous in the flask. Connect the parts of apparatus and flush air from the apparatus by turning on the hydrogen production for about 5 minutes. Then heat the solution. Bubbles of PH_3 will escape from the tube in the beaker and, on contact with air, will ignite.

To stop the flow gas, carefully fill the beaker with water and cease heating of the generator.

PH_3 - it is a colourless, highly toxic, flammable gas with a boiling point of -88°C at standard pressure . It can ignite spontaneously on contact with air. Pure phosphine is odourless, but "technical grade" phosphine has a highly unpleasant odour like garlic or rotting fish, due to the presence of substituted phosphine and diphosphine (P_2H_4).



Apparatus for PH_3 preparation

2.5.5 Reactions of phosphate ions (use 5 % Na_3PO_4 , NaH_2PO_4 , Na_2HPO_4 solutions)

a) Measure pH of all solutions.

b) Add some crystals of CH_3COONa (give reason) and 2-3 drops of 1 % AgNO_3 solution to 1 ml of solution of phosphate. Observe colour of the precipitate and test its solubility in $\text{NH}_3(\text{aq})$ and HNO_3 .

c) Repeat reaction (b), using 5 % FeCl_3 solution in place of AgNO_3 and test solubility of the precipitate in HCl (1:1) and HNO_3 .

d) Add a few drops of magnesium reagent (solution containing MgCl_2 , NH_4Cl , $\text{NH}_3(\text{aq})$) to 1 ml of phosphate solution.

e) Add a few drops of phosphate solution to 2 ml of ammonium molybdate reagent and heat it. Dissolve the precipitate in diluted NH_3 and then add diluted HNO_3 .

2.3.6 Preparation of sodium thiosulphate (8 g)



Material: $\text{Na}_2\text{SO}_3 \cdot 5\text{H}_2\text{O}$, S, ethanol

Procedure: Dissolve calculated amount of Na_2SO_3 in 30 ml of water. Add sulphur (10 % excess) moistened with ethanol and boil gently (about 30 min), add water if necessary, to keep the volume of the solution about the same level throughout boiling. Filter the mixture and evaporate the filtrate on a steam bath to the crystallisation. Dry the crystals by pressing between sheets of filter paper.

2.3.7 Reactions of thiosulphate ion (use 5 % $\text{Na}_2\text{S}_2\text{O}_3$ solution)

(record your observations and write the balanced equations for all reactions)

- Add a few drops of conc. HCl to 1–2 ml of $\text{Na}_2\text{S}_2\text{O}_3$ solution. Observe the changes and cautiously smell the gas which is evolved.
- Add dropwise $\text{Na}_2\text{S}_2\text{O}_3$ solution to potassium triiodide solution (solution of iodine in 10 % KI solution).
- Add 1 drop of 5 % $\text{Na}_2\text{S}_2\text{O}_3$ solution to 1 ml of 5 % AgNO_3 solution, in the other test tube add 1 drop of 5 % AgNO_3 solution to 1 ml of 5 % $\text{Na}_2\text{S}_2\text{O}_3$ solution. Compare the reactions in both test tubes.
- Add 5 % $\text{Na}_2\text{S}_2\text{O}_3$ solution to 1 ml of 5 % lead acetate solution and heat the suspension.

2.3.8 Qualitative reactions of SO_4^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, S^{2-}

Prepare 5 % solutions of salts of these anions. Transfer 1 ml of each of test solutions to the test tubes and perform reactions with following reagents: 5 % BaCl_2 , 5 % AgNO_3 , concentrated H_2SO_4 and solution of iodine in 10 % KI. Use new 1 ml of test solution for each reagent and stir after each addition. Record the results in table.

2.3.10 Preparation of hydrogen sulphide, $(\text{NH}_4)_2\text{S}$ and $(\text{NH}_4)_2\text{S}_x$ - perform this experiment in a fume hood

Material: FeS, HCl, $\text{NH}_3(\text{aq})$, S

Procedure:

- Prepare H_2S in a Kipp generator, fill a Kipp generator with sticks of FeS and put HCl (1:1) into the acid reservoir. Pass the gas into bottle with distilled water to prepare saturated solution of H_2S .
- $(\text{NH}_4)_2\text{S}$ – dilute $\text{NH}_3(\text{aq})$ with water (1:1) and divide it in two parts. Saturate one part with H_2S and mix it with the other one.
- $(\text{NH}_4)_2\text{S}_x$ – add excess of sulphur to the $(\text{NH}_4)_2\text{S}$ solution, filter the unreacted sulphur after 30 minutes.

3.2.4 Reactions of Al^{3+} (use $0.2 \text{ mol.l}^{-1} \text{ Al}_2(\text{SO}_4)_3$ or $\text{KAl}(\text{SO}_4)_2$ solution)
(record your observations and write the balanced equations for all reactions)

a) Pour 2 ml of Al^{3+} solution to the four separated test tubes and add these reagents (dropwise): diluted NH_3 , NaOH , Na_2CO_3 and Na_2S solutions. Compare the precipitates in all test tubes.

Dissolve the precipitate by adding a slight excess of NaOH solution, then add excess of NH_4Cl and heat the mixture.

b) Pour 2 ml of Al^{3+} solution to the four separated test tubes, add half a spoonful of tartaric acid and repeat reactions (a). Compare the results.

c) Add sufficient amount of NaOH solution to 1 ml of Al^{3+} solution to give $\text{Na}[\text{Al}(\text{OH})_4]$ solution and add alizarin reagent. Then drop acetic acid.

3.3.7 Reactions of Pb^{2+} (use $0.2 \text{ mol.l}^{-1} \text{ Pb}(\text{NO}_3)_2$ or $\text{Pb}(\text{CH}_3\text{COO})_2$ solution)

a) Pour 1-2 ml of Pb^{2+} solution in two test tubes add (dropwise) 5 % NaOH solution to one and 5 % $\text{NH}_3(\text{aq})$ to the other. Observe the reactions in the excess of reagents.

b) Add dilute HCl (1:3) dropwise to 2 ml of Pb^{2+} solution. Add 5 ml of distilled water to the solid precipitate, heat it in the boiling water bath. Then cool it on air. Observe the changes.

c) Add diluted HCl drop by drop to 1-2 ml of Pb^{2+} solution, pour off the solution above precipitate. Add more diluted HCl to the precipitate and dilute the obtained solution with water.

d) Add diluted H_2SO_4 (1:10) dropwise to 2 ml of Pb^{2+} solution. Centrifuge the precipitate and decant it. Put half of the precipitate into another test tube. Add 2-3 ml of concentrated H_2SO_4 to one tube and ammonium acetate with a few drops of $\text{NH}_3(\text{aq})$ to the other.

e) Add 10 % KI solution dropwise to 2 ml of Pb^{2+} solution. Then add distilled water to the solid precipitate, heat it in the boiling water bath. Then cool it on the air.

f) Add 5% K_2CrO_4 solution to 2 ml of Pb^{2+} solution.

g) Add H_2S to 2 ml of Pb^{2+} solution, boil the precipitate with hydrogen peroxide (3 %). Add H_2S to 2 ml of Pb^{2+} solution in the second test tube and then add saturated KCl solution.

4.4.5 Reactions of Fe^{3+} (use 0.5 mol.l^{-1} solution of Fe^{3+} salt)

a) Put 1 ml of Fe^{3+} solution in each of seven test tubes (or on a spot plate). Add a few drops of the following seven reagents: 1 % NH_4SCN , 5 % $\text{K}_3[\text{Fe}(\text{CN})_6]$, 5 % $\text{K}_4[\text{Fe}(\text{CN})_6]$, 5 % NaH_2PO_4 , 5 % NaOH , 5 % KI and $\text{NH}_3(\text{aq})$ to each of the seven test tubes. Observe reactions and record them.

b) Add 5 % KF solution and 1 % NH_4SCN solution to 1 ml of Fe^{3+} solution. Explain your observation.

4.4.6 Reactions of Fe^{2+} (use 0.5 mol.l⁻¹ solution of fresh prepared Fe^{2+} salt)

- a) Repeat reactions 4.4.5 a) using Fe^{2+} solution in place of Fe^{3+} solution.
- b) Add one drop of H_2SO_4 (1:2) and a few drops of KMnO_4 solution to 1 ml of Fe^{2+} solution. Then add a few drops of NH_4SCN solution.

4.3.3 Reactions of Mn^{2+} (use 0.3 mol.l⁻¹ solution of Mn^{2+} salt)

- a) Add $\text{NH}_3(\text{aq})$ to 2 ml of Mn^{2+} solution. Observe reaction of precipitate with excess of ammonia. Then add 1-2 drops of H_2O_2 .
- b) Use very diluted Mn^{2+} solution (0.02 mol.l⁻¹), acidify it with HNO_3 and add solid $\text{K}_2\text{S}_2\text{O}_8$ and one drop of 1 % AgNO_3 solution. Heat the mixture. Compare with reaction without AgNO_3 .
- c) Add KMnO_4 solution dropwise to 1 ml of Mn^{2+} solution and observe changes of colour.

4.5.11 Reactions of Cu^{2+} (use 0.3 mol.l⁻¹ solution of Cu^{2+} salt, test tubes or a spot plate)

- a) Add about 0.5 ml of $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution to 1-2 ml of Cu^{2+} solution. Try the solubility of precipitate in HCl , CH_3COOH and $\text{NH}_3(\text{aq})$.
- b) Repeat the reaction (a) with 5 % $\text{K}_3[\text{Fe}(\text{CN})_6]$ solution.
- c) Add 10 % NaOH solution to 1-2 ml of Cu^{2+} solution to complete the precipitation. Heat the mixture and observe change of colour.
- d) Add 2 ml of tartaric acid solution to 1-2 ml of Cu^{2+} solution and repeat experiment (c). Explain the difference.
- e) Add concentrated $\text{NH}_3(\text{aq})$ (a few drops at a time) to 1-2 ml of Cu^{2+} solution. Record your observation. Add ammonia solution until there is no further change. Then add two pellets of NaOH and boil.
- f) Add a few drops of 5 % KI solution to 1-2 ml of Cu^{2+} solution. Add a few drops of Na_2SO_3 solution to determine colour of the precipitate.
Precipitation of CuI , which is white, but the solution is brown because of formation of tri-iodide ions (iodine). After adding Na_2SO_3 to the solution, tri-iodide is reduced to colourless iodide ions and the white colour of the precipitate becomes visible.
- g) Add concentrated HCl dropwise to 1-2 ml of Cu^{2+} solution and observe change of solution. Dilute the solution with water.

4.4.11 Reactions of Ni^{2+} (use 0.5 mol.l^{-1} solution of Ni^{2+} salt, test tubes or a spot plate)

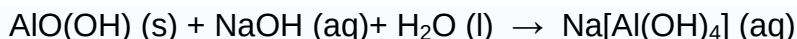
a) Add diluted NH_3 and 0.5 ml of dimethylglyoxime reagent to 2 ml of Ni^{2+} solution. The Ni^{2+} is first complexed with ammonia and then detected as an insoluble, scarlet coordination compound of dimethylglyoxime (DMGH_2).

b) Add NaOH solution to 2 ml of Ni^{2+} solution to complete the precipitation. Divide the precipitate into two test tubes. Add excess of NaOH to one and a few drops of H_2O_2 to the other.

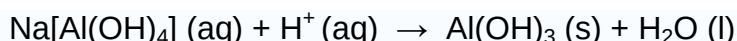
c) Add $\text{NH}_3(\text{aq})$ dropwise to 2 ml of Ni^{2+} solution. Try the action of $\text{NH}_3(\text{aq})$ excess. Then add NaOH solution.

3.2.2 Preparation of Al_2O_3 from bauxite

Bauxite, $\text{AlO}(\text{OH})$, the most important ore of aluminum, contains only 40-60% of Al_2O_3 , the rest being a mixture of silicon dioxide and various iron oxides. The aluminum oxide must be purified before it can be refined to aluminum metal. Bauxite is washed with a hot solution of sodium hydroxide. This converts it to aluminum hydroxide, $\text{Al}(\text{OH})_3$, which dissolves in the hydroxide solution according to the chemical equation



The other components of bauxite do not dissolve and can be filtered out as solid impurities. Then do precipitation of aluminum hydroxide by carbon dioxide, CO_2 . $\text{Al}(\text{OH})_3$ is then filtered and dried. When then heated, the aluminum hydroxide decomposes to alumina, giving off water vapour in the process:



Material: bauxite, NaOH , CO_2 , CH_3COOH

Procedure: Calculate amount of NaOH (10 % excess) for reaction of 10 g of bauxite and prepare 30 % NaOH solution. Leach 10 g of bauxite in this solution for 20 minutes, adding water if necessary, to keep the volume about the same level throughout leaching. Dilute the suspension to a triple volume. Filter the insoluble component ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$). Use suitable method of filtration of strongly basic solution. Divide the filtrate in two parts, the first one precipitate with CO_2 (wash bottle) and the second one with diluted CH_3COOH . Filter out the precipitates (separately) and wash them with water until the filtrate is not acid (second part).

Put the precipitates (separately) into crucible and heat gently at first and then strongly. Try the action of diluted and concentrated acids and hydroxide on the product before and after its heating.

2.6.5 Reduction effect of oxalic acid

A particular case of catalysis is so-called autocatalysis. Here, a slowly-reacting system itself produces a positive catalyst. As a result, the original reaction runs faster and faster. The reaction of permanganate with oxalate is a well-known example in which the Mn^{2+} ions produced by the reduction of MnO_4^- catalyse the reduction of further permanganate

a) Put 1 ml of KMnO_4 solution to the three test tubes. Add 1-2 drops of H_2SO_4 to the first and second tube. Add 1 drop of Mn^{2+} to the second tube. Add excess of $\text{H}_2\text{C}_2\text{O}_4$ solution to all three test tubes. Observe rate of reactions and write the equations.

2.6.2 Reactions of carbon dioxide and carbonates

- a) Put 3 ml of 5 % BaCl_2 solution in two separate test tubes. Add 1-2 drops of NaOH solution to one and pass CO_2 to each. When the precipitate appears, heat the test tube.
- b) Repeat experiment (a) using 1 % AgNO_3 solution in place of BaCl_2 .
- c) Add 1 drop of phenolphthalein to 2 ml of 1 % Na_2CO_3 solution and pass CO_2 to decolourize the solution. Explain this reaction.
- d) Add about 5 ml of water to the half a spoonful of solid CaCO_3 and shake the mixture. Observe solubility. Pass CO_2 and observe the change, then boil it.

2.6.6 Temperature stability of carbonates

Material: portion again. Order the carbonates according to their temperature stability. MgCO_3 , CaCO_3 , BaCO_3

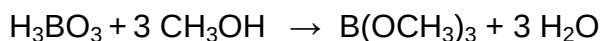
Procedure: Weigh three crucibles and put 3 g of MgCO_3 , CaCO_3 , BaCO_3 there. Dry the samples in a drying box at 105°C to the constant mass. Calculate humidity in the samples. Heat the crucibles by Mecker burner (or in electrical furnace) for 5 minutes. Allow the crucibles to cool down to room temperature and weigh them. Handle the crucibles with crucible tongs always. Calculate amount (%) of carbonates decomposed by heat. Heat again for another 15 minutes and calculate the decomposed

3.1.8 Reactions of magnesium, calcium, strontium and barium ions

(observe when the precipitate appears, prepare a table for recording the results of these experiments)

- a) Prepare 0.1 mol.l^{-1} solutions of Mg^{2+} , Ca^{2+} , Sr^{2+} , Ba^{2+} salts (chlorides or nitrates).
- b) Add 1 M Na_2CO_3 solution drop by drop to 1–2 ml of each salt solution. Repeat this reaction with 1 M $(\text{NH}_4)_2\text{CO}_3$ solution.
- c) Add diluted H_2SO_4 (1:3) to 1–2 ml of each solution. Determine whether or not the **formed** sulphate will react with hydrochloric acid.
- d) Repeat reaction (c) using 0.5 M $\text{H}_2\text{C}_2\text{O}_4$ and K_2CrO_4 solutions in place of H_2SO_4 .

2.8.1 Preparation of methyl borate



Material: H_3BO_3 , CH_3OH , H_2SO_4

Procedure: Put 0.5 g of dry H_3BO_3 crystals into evaporating dish, add 10 ml of methanol, 1 ml of concentrated H_2SO_4 and ignite the solution. What colour is the flame? Explain the role of H_2SO_4 .

3.1.1 Flame tests

Compounds of certain metals are volatilized in a non-luminous Bunsen flame and impart characteristic colours to the flame. Chlorides are among the most volatile compounds, and these are prepared in situ by mixing the compound with a little of concentrated HCl before carrying out the tests.

Material: alkaline metals and earth salts (solid or solution), HCl, platinum wire

Procedure: Clean a platinum wire by dipping it into concentrated HCl in the test tube and holding it in the oxidizing zone of Bunsen flame (never insert the wire into reducing zone of flame!). The wire is clean when it imparts no colour to the flame. The wire is dipped into the solution of salt (or into HCl and then into crystals of salt) and then hold in the flame.

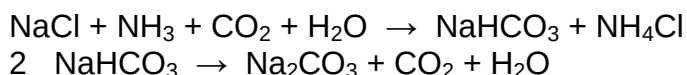
3.1.2 Insoluble salts of alkaline metals (*use 1 M solutions of salts*)

a) Add 1 ml of 5 % Na_2HPO_4 solution and $\text{NH}_3(\text{aq})$ to 1 ml of Li^+ solution. What is the colour of the precipitate? Try the action of HCl on the precipitate.

b) Add a few drops of 10 % $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ solution to each of three test tubes containing 1 ml of solutions of Na^+ , K^+ and NH_4^+ salts. Observe which ion precipitates and try solubility of the precipitate in HCl.

c) Add a few drops of HClO_4 to 1 ml of each solutions of Na^+ and K^+ salts. Describe the appearance of the precipitate.

3.1.3 Preparation of sodium carbonate



Material: NaCl, $\text{NH}_3(\text{aq})$, $\text{C}_2\text{H}_5\text{OH}$, CO_2

Procedure: Saturate 40 ml of conc. $\text{NH}_3(\text{aq})$ with 10 g of NaCl in a wash bottle. Shake the mixture vigorously to dissolve all salt. Pass the CO_2 until a heavy precipitate of NaHCO_3 is obtained. Filter out the precipitate, wash it with 5 ml of ice water and then with 10 ml of $\text{C}_2\text{H}_5\text{OH}$. Dry NaHCO_3 at 100°C , put it into dish and heat gently (300°C).

3.3.6 Preparation of Pb_3O_4 -“red lead” (5 g)



Material: PbCO_3 (PbO), NaNO_3 , KClO_3

Procedure: Mix well together calculated amounts of PbCO_3 and NaNO_3 . Stir and heat it gently in the porcelain crucible. First decomposition of PbCO_3 occurs at 200°C to yellow PbO and CO_2 . When all mixture is yellow increase the temperature to about $400 - 450^\circ\text{C}$ (crucible is not red) and stir continually. When the mixture turns red add 1-2 g KClO_3 , stir and heat it for 30 minutes. When the temperature is higher (470°C) Pb_3O_4 decomposes.

3.3.8 Reactions of Pb_3O_4

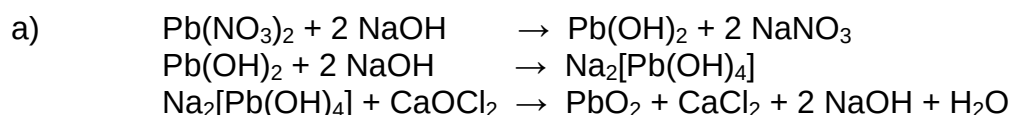
(record your observations and write the balanced equations for all reactions)

a) Add 3 ml of diluted HNO_3 (1:1) to a small bit of Pb_3O_4 in a test tube. When no further reaction takes place, filter the liquid. Add diluted HCl to the residue and detect the evolved gas with a wet starch iodide paper. Add a few drops of sulphuric acid or KI solution to the filtrate. What do you detect with this reaction?

b) Warm a small bit of Pb_3O_4 in a test tube with 2-3 ml of concentrated HCl . Detect the evolved gas with a wet starch iodide paper.

c) Place a small amount of Pb_3O_4 into a crucible and heat it strongly. Observe change of colour. Cool the product and repeat the reaction (a).

3.3.5 Preparation of lead dioxide



Material: $\text{Pb(NO}_3)_2$ or $\text{Pb(CH}_3\text{COO)}_2$, NaOH , CaOCl_2 or NaClO , HNO_3

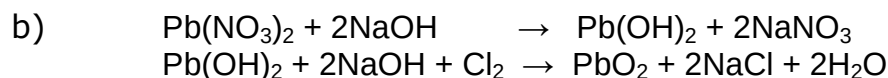
Procedure: Dissolve 0.1 mol of Pb^{2+} salt in 300 ml of hot water and cool the solution. If a small amount of substance remains undissolved, it will not affect the result. Add the solution of 20 g NaOH in 180 ml of water. First white precipitate of Pb(OH)_2 appears, which dissolves in excess of NaOH to $\text{Na}_2[\text{Pb(OH)}_4]$.

Mix 40 g of CaOCl_2 with 50 ml of water and necessary amount of Na_2CO_3 for reaction with Ca^{2+} . Add 200 ml of water to the mixture and filter it.

Add filtrate to the boiling solution of $\text{Na}_2[\text{Pb(OH)}_4]$ until test for Pb^{2+} is negative.

Test for Pb^{2+} : add one drop of Na_2S solution to one drop of reaction mixture on filter paper. Black precipitate (PbS) indicates presence of Pb^{2+} .

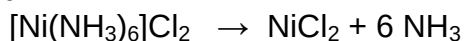
Pour the mixture into 500 ml of water and decant. Add 100 ml of diluted HNO_3 (1:3) to the precipitate, stir and decant it until pH is neutral. Filter off the lead dioxide, wash it with boiling water and dry.



Material: $\text{Pb(NO}_3)_2$ or $\text{Pb(CH}_3\text{COO)}_2$, NaOH , Cl_2

Procedure: Dissolve 10.0 g of lead(II) nitrate in 80 ml of water acidified with a drop of concentrated nitric acid and add solution of 2.4 g of sodium hydroxide dissolved in 50 ml of water slightly with constant stirring. Heat precipitated lead(II) hydroxide suspension to 70 - 80 °C and bubble with chlorine at the same time (do not heat it over 80 °C or PbO will be obtained!). Decant the precipitated brown-black product with diluted nitric acid (1:3) and then with water. Dry it at 100 °C.

4.4.10 Preparation of anhydrous nickel chloride - perform this experiment in a fume hood

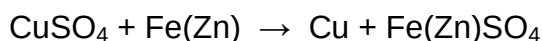


Material: NiCO_3 , HCl , $\text{NH}_3(\text{aq})$

Procedure: Dissolve 8 g of NiCO_3 in calculated amount of diluted HCl (1:1). About 5 % of NiCO_3 has to stay undissolved. Heat to expel CO_2 , add water if necessary to keep the volume about the same level throughout heating. Filter it, add 40 ml of concentrated $\text{NH}_3(\text{aq})$ in small amounts to the filtrate and stir it. Primarily appearing $\text{Ni}(\text{OH})_2$ precipitate has to dissolve to a complex. Filter if necessary and add solution of 10 g of NH_4Cl in 25 ml of water.

Filter off the $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$ on a Büchner funnel and wash it with solution of 10 g of NH_4Cl in 50 ml of $\text{NH}_3(\text{aq})$. Put the moisten product on evaporation dish, heat it gently and stir. Observe change of colour and test the evolved gas.

4.5.2 Preparation of copper



Material: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, Fe (nails) or Zn, H_2SO_4 , HCl

Procedure: Prepare (in a beaker) 10 % solution from 4 g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and add a few drops of concentrated H_2SO_4 . Place the beaker on wire gauze, heat it gently to 50 °C and add iron nails or Zn (in 20 % excess). Heat and stir it until the colour of solution will be light green. Filter off the copper.

The excess of Zn will be dissolved from the product by means of diluted HCl , which does not affect the copper. Add in small portions 10 ml of HCl (1:1) in about 5 minutes. Heat it gently and stir to determine if hydrogen is still evolved from the excess of Zn used. Filter the copper out and wash it with water. Store the copper under the water.

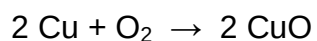
4.5.6 Preparation of copper ammine sulphate - perform this experiment in a fume hood



Material: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{NH}_3(\text{aq})$, $\text{C}_2\text{H}_5\text{OH}$

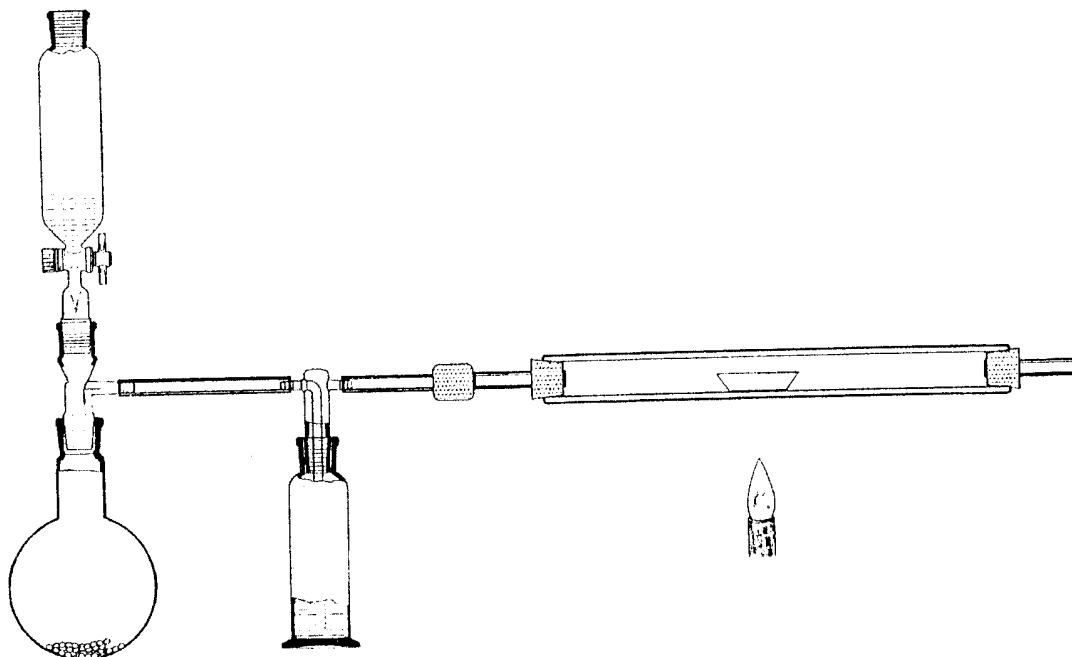
Procedure: Place 5 g of finely powdered copper sulphate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, in a small beaker, pour upon it 7.5 ml of concentrated ammonia and 3 ml of water. Shake for about 1 minute and then heat it gently until all the solid dissolves. Add about 10 ml of ethanol to the solution, let it stand for about one hour and filter off the crystals. Wash them with mixture of 5 ml of concentrated ammonia and 5 ml of ethanol. Dry them on air in the hood.

4.5.3 Preparation of copper oxide



Material: Cu (from experiment 4.5.2), for oxygen preparation: H_2O_2 , MnO_2

Procedure: Set up a gas production apparatus (see 2.1.2) and combustion tube (see figure), where a ceramic boat with Cu is placed. Pass the oxygen through wash bottle with concentrated H_2SO_4 and heat the tube under the boat for 20-30 minutes. Discontinue the heating, but continue passing oxygen through the apparatus until the combustion tube is sufficiently cool to be touched by the hand.



Apparatus for oxidation (reduction) by gas

Qualitative analysis

Qualitative analysis schemes are used to help students understand reactivity and to develop problem solving skills. There are several different approaches to these experiments. In one case, students are given a step-by-step procedure, often in the form of a flow chart, which they can use to isolate and identify unknown ions in solution. In another, students first analyze known solutions to determine how different ions will behave when reacted with various reagents. They then compile these results into their own flow chart that they apply to their unknown samples. The experiments can be carried out on solutions containing mixtures of cations (same anion), mixtures of anions (same cation) or on salt mixtures.

The separation flow chart for the cations and anions are shown in figures. These flow charts show the steps required to separate and identify the cations and anions that you may find in your known and unknown mixtures. These charts have been prepared based on theoretical information about the ions and experimental observations. The flow charts can help you understand the order in which separations must take place in order to isolate ions that may behave similarly.

There are several procedural steps that are indicated in the flow chart that are described here.

Precipitation: after addition of a precipitating agent, it is important to mix the solution thoroughly by shaking or by stirring with a clean glass rod. Be sure not to add more of a precipitating agent than indicated in the procedure as this may cause undesired side reactions.

Separation: after addition of a precipitating agent to a solution, mixture of a solid (precipitate) and liquid (supernatant) phase will be obtained. These often must be isolated and treated separately. The most common method for separating of a supernatant and a precipitate is to centrifuge the mixture in order to gather the solid at the bottom of the test tube. It is essential that the centrifuge is balanced with a test tube containing the same volume of liquid as your sample. After centrifuging, the supernatant can be decanted by simply pouring from one test tube into another or by careful removal with a clean glass pipette.

Washing: when a supernatant is removed from a solid, it is almost certain that some of the liquid has been left behind. This liquid can be removed by the addition of a clean solvent (usually cold water, but indicated in the procedure if not), which is thoroughly mixed with the precipitate. After centrifuging and decanting, the solid is now ready for further reaction as dictated in the procedure.

Heating of the solutions: there are two different heating methods in qualitative analysis. If a solution needs to maintain at a near boiling temperature for a few minutes, the test tube should be placed in a boiling water bath. If the solution needs to be heated directly, the test tube should be held in a test tube clamp facing away from you and your fellow students and passed back and forth through the flame of a Bunsen burner. Be careful not to let the solution bump and jump out of the test tube by keeping the flame near the surface of the

solution rather than at the bottom. Stirring of the solution with a glass rod may also help.

Flame tests - in many cases, the colour imparted to a flame when a cation is heated directly in a flame can help to identify the element. If a procedure requires a flame test, follow the directions below.

1) Clean a wire loop first by dipping it in 6 M HCl and then heating it long enough to drive off any contaminants from the surface.

2) Place the loop in the solution or solid to be tested, making sure a drop of liquid or a crystal remains in the loop.

3) Place the wire in the flame and observe the imparted colour. If instructed, view the flame through a piece of cobalt blue glass.

4.7.2 Qualitative analysis of cations from group I – III

Metal ions can be classified according to their behaviour towards certain group reagents. This type of classification divides the elements into analytical groups. The basic principle of this separation process is that of adding anions consecutively to the acid solution of cations, so that the greatest number of metal ions is precipitated together. These precipitants are the hydrochloric acid group, the hydrogen sulphide group, the ammonium sulphide group and the ammonium carbonate group.

Group reagent	H ₂ S			(NH ₄) ₂ S		(NH ₄) ₂ CO ₃	
	HCl	H ₂ S (H ⁺)					
Group	I.a	I.b	II.	III.a	III.b	IV.	V.
Cations	Ag ⁺	Cu ²⁺	As ³⁺	Al ³⁺	Mn ²⁺	Ca ²⁺	Mg ²⁺
	Pb ²⁺	Hg ²⁺	As ⁵⁺	Cr ³⁺	Ni ²⁺	Sr ²⁺	Na ⁺
	Hg ₂ ²⁺	Bi ³⁺	Sb ³⁺	(Fe ³⁺)	Fe ²⁺	Ba ²⁺	K ⁺
	Tl ⁺	Cd ²⁺	Sb ⁵⁺		Fe ³⁺		NH ₄ ⁺
			Sn ²⁺		Co ²⁺		H ⁺
			Sn ⁴⁺		Zn ²⁺		

Material: 5 % solutions of cations from group I – III, HCl, H₂S (gas or saturated aqueous solution), (NH₄)₂S, (NH₄)₂S_x.

Procedure: Use 1-2 ml of solution of each cation for reaction, observe colour of solutions, adjust pH of each solution. Tabulate the results (appearance, colour and solubility of precipitates).

a) Add HCl (1:4) dropwise to 1 ml of the solutions of Ag⁺ and Pb²⁺ salts in the separated test tubes. When the precipitate settles add an additional drop or two of HCl and let the solution sit, to see if more AgCl or PbCl₂ precipitate forms. Be careful to avoid a large excess of HCl, since both cations form soluble complex cations with excess chloride ions.

b) Remove the precipitates from reaction (a) by centrifugation (or decantation). Add ~20 drops of hot distilled water to the solid precipitates and observe their solubility.

c) If necessary add HCl (1:4) to solutions of cations from group I.b and II in separated test tubes (confirm acidity with pH paper), and then add $\text{H}_2\text{S}(\text{aq})$. Observe colour of the precipitates.

d) Remove the precipitates of sulphides from reaction (c) by centrifugation (or decantation) and add 0.5 ml of concentrated HNO_3 . Observe if precipitates dissolve.

e) Separate the sulphide precipitates of cations from group II by centrifugation and add $(\text{NH}_4)_2\text{S}_x$ solution, shake and heat it. Observe dissolving of precipitates (formation of thio-complexes). Add acetic acid (1:1) to solution of thio-complex to the acid reaction (check pH with an indicator paper) and heat it. Describe the reaction.

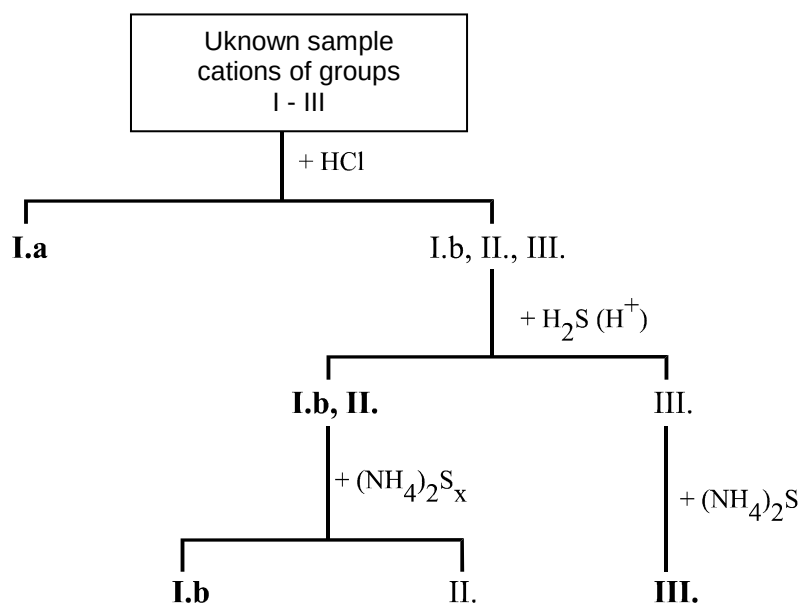
f) Add a few drops of $(\text{NH}_4)_2\text{S}$ solution to solutions of cations from group III in separated test tubes. Observe colour of the precipitates.

g) Prepare mixture with one cation from each of group I – III into the test tube and separate the cations with help of group reagents.

4.7.3 Determination of cations in unknown samples

The samples contain just one or mixture of two cations from different analytical groups (I-III). Determine cations by reactions of group reagents (4.7.2) and by colour of the precipitates. Note any simple physical characteristics of the samples such as color, pH (use indicator paper), etc.

Use flowchart and your observations to confirm presence or absence of each cation. To confirm your analysis of the unknown sample, use the known solutions of single cations to observe positive confirmation tests.



Flow chart for the qualitative analysis of cations from groups I-III

2.9.2 Determination of anions in unknown samples

a) Preliminary tests

- determination of pH of the sample
- determination of anions of volatile acid (reaction with H_2SO_4)

b) Determination of oxidation or reduction properties of anions

d) Precipitation reaction with Ba^{2+}

e) Precipitation reaction with Ag^+

Group reagent	H_2SO_4 (1:5)	$\text{Ba}(\text{NO}_3)_2$	AgNO_3	
Group	I.	II.	III.	IV.
Anion	CO_3^{2-} $\text{NO}_2^-(\text{t})$ SO_3^{2-} S^{2-} S_x^{2-} $\text{S}_2\text{O}_3^{2-}$ SiO_3^{2-}	SO_4^{2-} PO_4^{3-} BO_3^{3-} F^-	Cl^- Br^- I^- CN^- SCN^- $[\text{Fe}(\text{CN})_6]^{4-}$ $[\text{Fe}(\text{CN})_6]^{3-}$	NO_3^- NO_2^- CH_3COO^-

Flowchart for anions determination

Preliminary tests

Determination of pH of sample solution (indicator paper)

+ H_2SO_4 (1:1) - CO_3^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, S^{2-} , NO_2^{2-}

+ conc. H_2SO_4 - CO_3^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, S^{2-} , NO_2^{2-} and oxidation of Br^- , I^-

Redox reactions

Reducible anions:

+ KI (0.1 mol.l⁻¹ KI , 5 mol.l⁻¹ HCl)

O_2^{2-} , NO_2^- , $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^- , NO_3^- (more acidic solution)

Oxidable anions:

+ KMnO_4 (0.02 mol.l⁻¹ KMnO_4 , 2.5 mol.l⁻¹ H_2SO_4)

Br^- , I^- , O_2^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, S^{2-} , NO_2^- , SCN^- , $\text{C}_2\text{O}_4^{2-}$, Cl^- (more acidic solution)

+ I_2 (0.05 mol.l⁻¹ KI_3 , 5 mol.l⁻¹ HCl)

SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, S^{2-}

Precipitation reactions

+ Ba^{2+} (0.1 mol.l⁻¹ $\text{Ba}(\text{NO}_3)_2$, BaCl_2)

- insoluble in diluted CH_3COOH , HCl , HNO_3

SO_4^{2-} , SiF_6^{2-} , IO_3^-

- insoluble in diluted CH_3COOH , soluble or decomposing in diluted HCl

SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, F^- , $\text{C}_2\text{O}_4^{2-}$, CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$

- soluble or decomposing in diluted CH_3COOH

CO_3^{2-} , PO_4^{3-}

- + Ag^+ (0.1 mol.l⁻¹ AgNO_3)
 - soluble or decomposing in diluted HNO_3
 CO_3^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, PO_4^{3-} , CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$
 - insoluble in diluted HNO_3
 - insoluble in concentrated $\text{NH}_3(\text{aq})$
 I^- , S^{2-} , S_x^{2-}
 - insoluble in diluted $\text{NH}_3(\text{aq})$, soluble in concentrated $\text{NH}_3(\text{aq})$
 Br^- , SCN^-
 - soluble in diluted $\text{NH}_3(\text{aq})$
 Cl^- , CN^-
 - anions do not precipitate with Ba^{2+} and Ag^+
 NO_2^- , NO_3^- , ClO_3^- , ClO_4^- , MnO_4^-

4.7.4 Procedure for analysis of cations from group IV

The ions remaining in solution after the separation of cations from group III are those of the alkaline earths, the alkali metals, and the ammonium.

Because substances containing sodium or ammonium ions are used in the general separation scheme, tests for the presence of Na^+ and NH_4^+ are carried out on an original sample of solution. Potassium ion is also identified by a flame test made on the original sample.

Step 1.

If you are working with only the group IV cations start with step 2. If you are analyzing a general unknown solution (solution remaining after removal of the group III cations), boil the solution down to a volume of 2 ml. Transfer the liquid to a test tube and centrifuge out any solid matter, which you may discard. Add 1 ml of 6 M HCl to the liquid in the beaker, and boil it to dryness (to drive off all the ammonium salts that were added in previous steps). Let the beaker cool down, and then add 0.5 ml of 6 M HCl and 2 ml of water. Warm the solution gently to dissolve the remaining salts.

Step 2.

Add 6 M NH_3 until the solution becomes basic, add further 0.5 ml. Then add 1 ml of 1M $(\text{NH}_4)_2\text{CO}_3$, stir and heat it in the water bath. Centrifuge it and decant the liquid (K^+ , Mg^{2+}). Wash the precipitate.

Step 3.

Confirmation of presence of Ba^{2+} . Add 0.5 ml of 6 M CH_3COOH to the precipitate (step 2) and stir it to dissolve the solid. Add 1 ml of H_2O and 2 drops of 6 M NH_3 and mix it, then add 0.5 ml of 1 M K_2CrO_4 . A yellow precipitate indicates presence of barium. Centrifuge out the solid phase and decant the yellow liquid.

Step 4.

Confirmation of presence of Ca^{2+} . Add 0.5 ml of 1 M $\text{K}_2\text{C}_2\text{O}_4$ to the liquid (step 3), stir it and let the solution stand for a minute. White precipitate is confirmation for presence of calcium. Centrifuge out the white solid and pour the liquid (Mg^{2+}) into the test tube.

Step 5.

Confirmation of presence of Mg^{2+} . To the solution (step 4) add 0.5 ml of 1 M Na_2HPO_4 , stir it and warm gently in the water bath. A white precipitate is indicative of presence of magnesium.

Step 6.

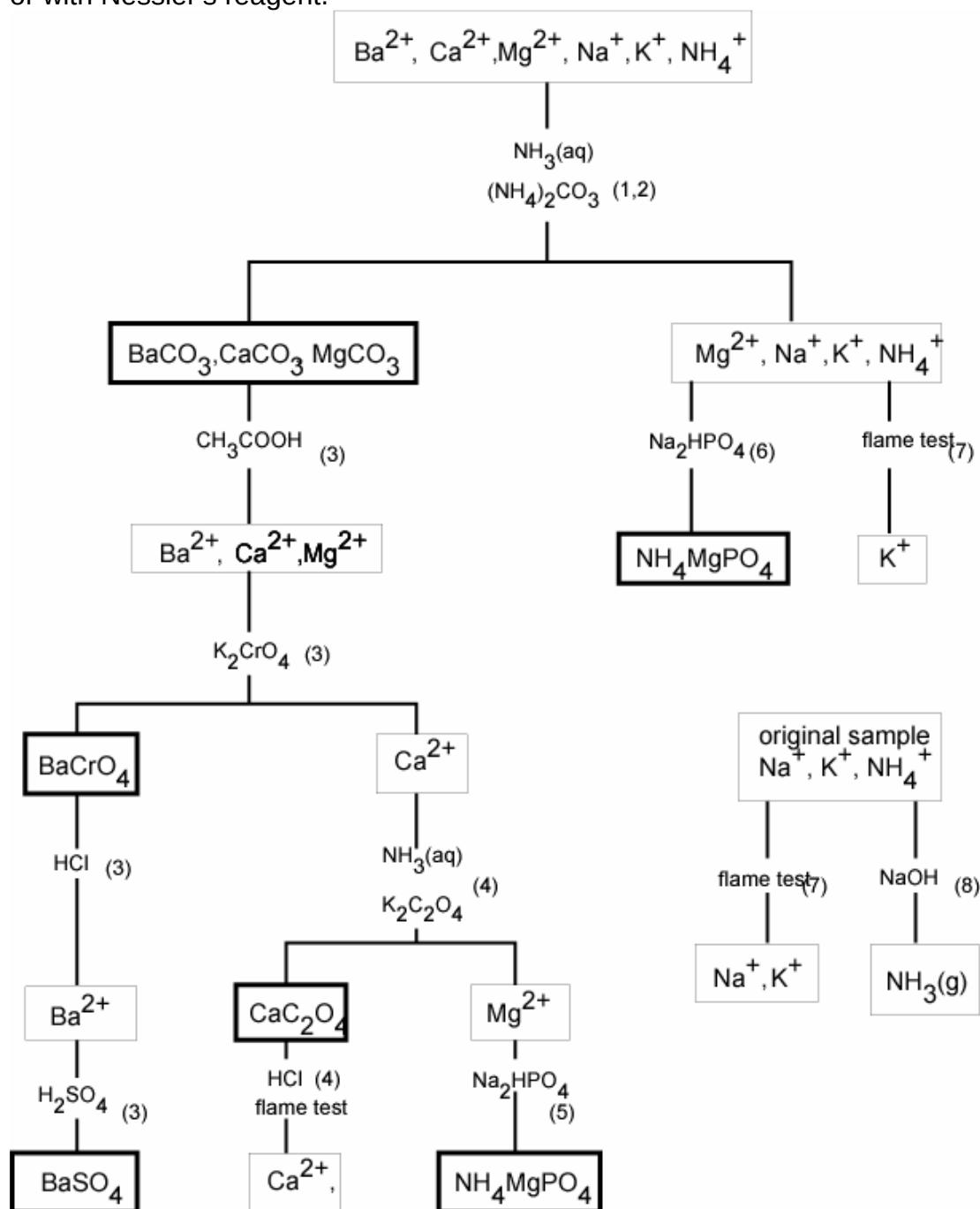
Alternative Confirmation of presence of Mg^{2+} . Repeat step 5 on the half of the solution obtained in step 2.

Step 7.

Confirmation of presence of Na^+ and K^+ . Perform a flame test on 1 ml of the original sample.

Step 8.

Confirmation of presence of NH_4^+ . Add 1 ml of 6 M NaOH to 1 ml of the original sample and heat it gently. Detect the ammonium with moisten piece of indicator paper or with Nessler's reagent.

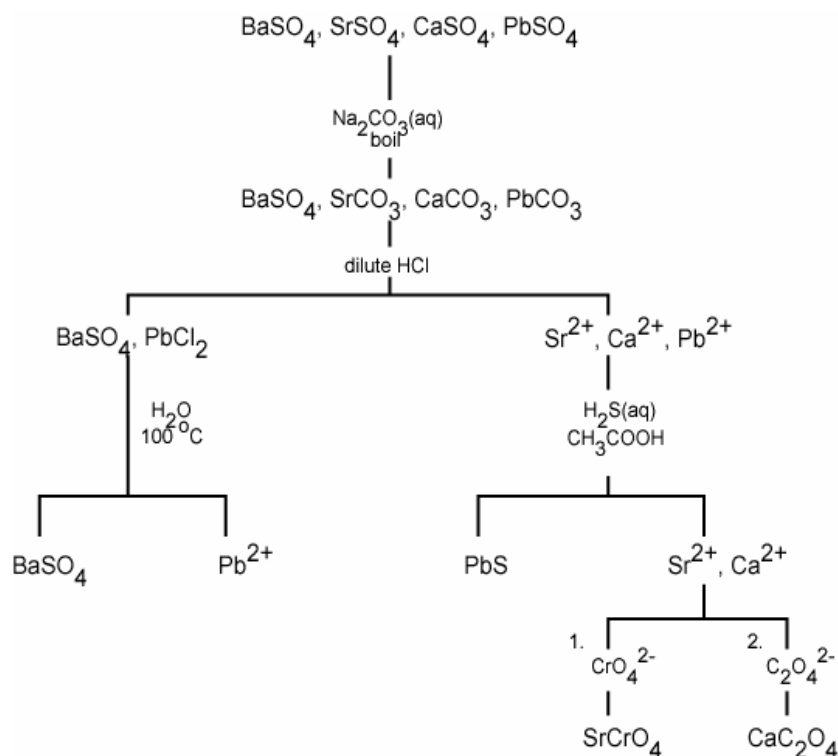


Flow chart for the qualitative analysis of cations from group IV

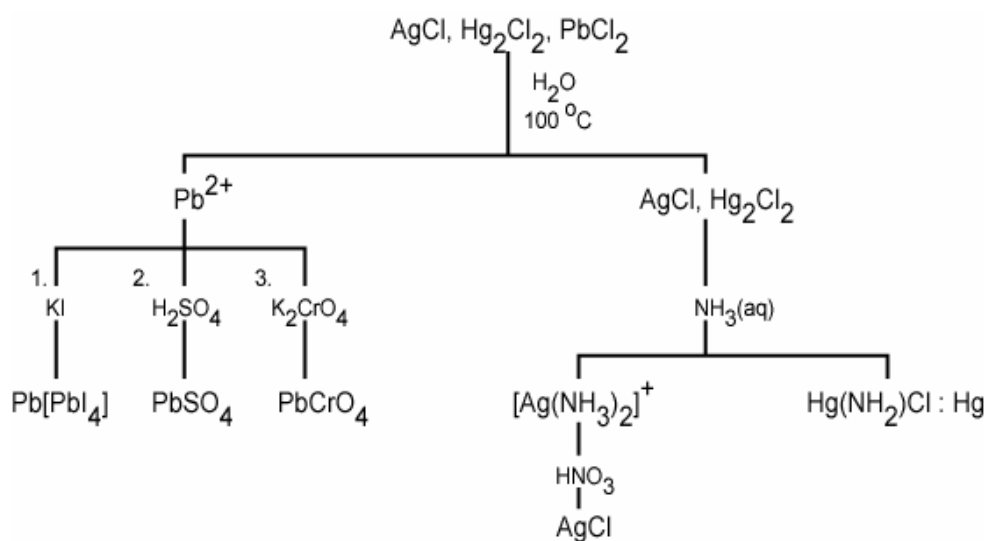
4.7.5 Separation and determination of cations of insoluble chlorides and sulphates groups

1. group of insoluble sulphates - Pb^{2+} , Ba^{2+} , Ca^{2+} , Sr^{2+}
2. group of insoluble chlorides - Ag^+ , Hg_2^{2+} , Pb^{2+}

After separation of all cations by precipitation with HCl (1:4) or H_2SO_4 (1:3) perform separation into group.



Flow chart for separation and determination of cations of insoluble sulphate group

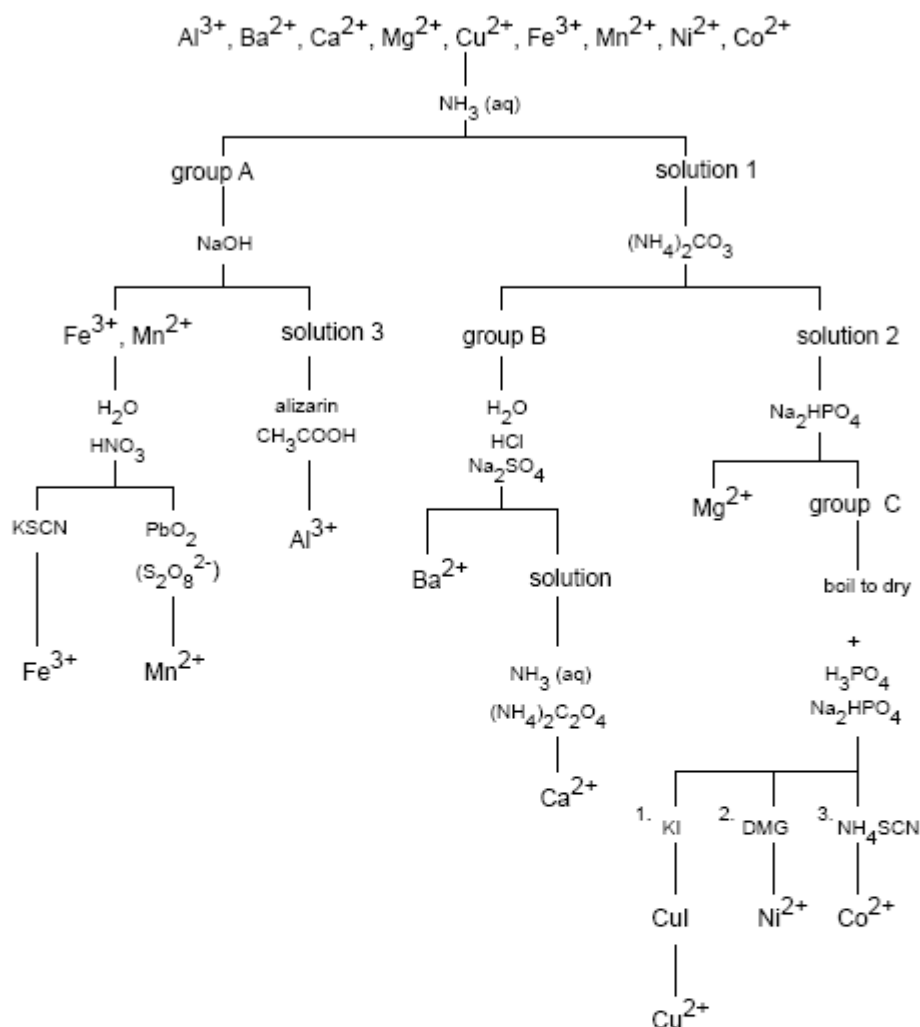


Flow chart for separation and determination of cations of insoluble chloride group

4.7.6 Separation and determination of cations from groups III and IV (without use H_2S)

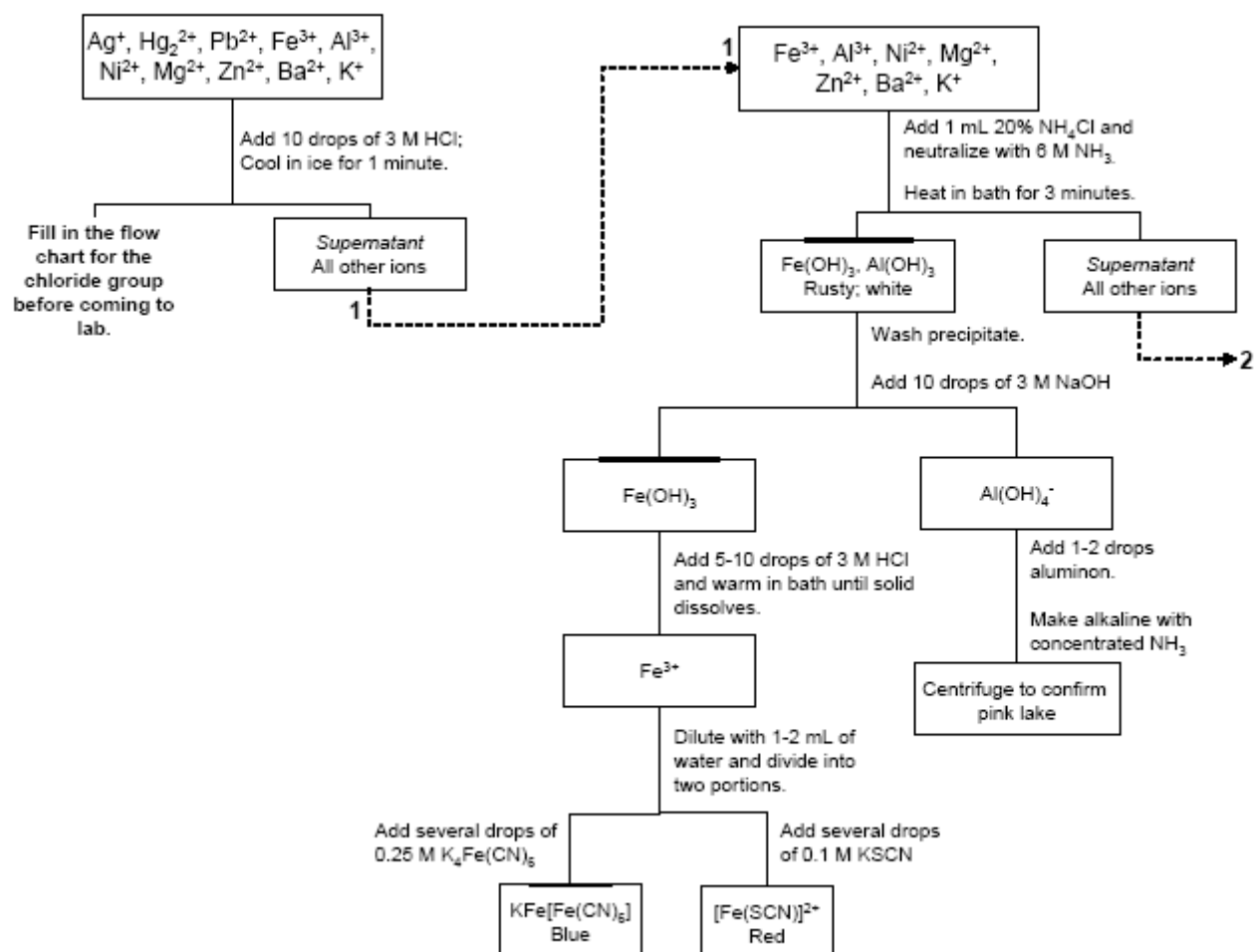
1. Add 1 ml of 6 M NH_3 to the 1 ml of the sample, stir it and centrifuge out the precipitate – group A (Al^{3+} , Fe^{3+} , Mn^{2+}).
2. Add 2 ml of 1 M $(\text{NH}_4)_2\text{CO}_3$ to the solution from (1), stir it and centrifuge out the precipitate – group B (Ca^{2+} , Ba^{2+}).
3. Add 1 ml of 1 M Na_2HPO_4 (confirmation of Mg^{2+}) to the solution from (2). Centrifuge the precipitate.
4. The remaining solution (3) contents group C (Co^{2+} , Ni^{2+} , Cu^{2+}).

Use known reactions and the flow chart for confirmation of particular cations.

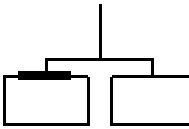
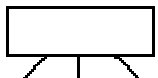


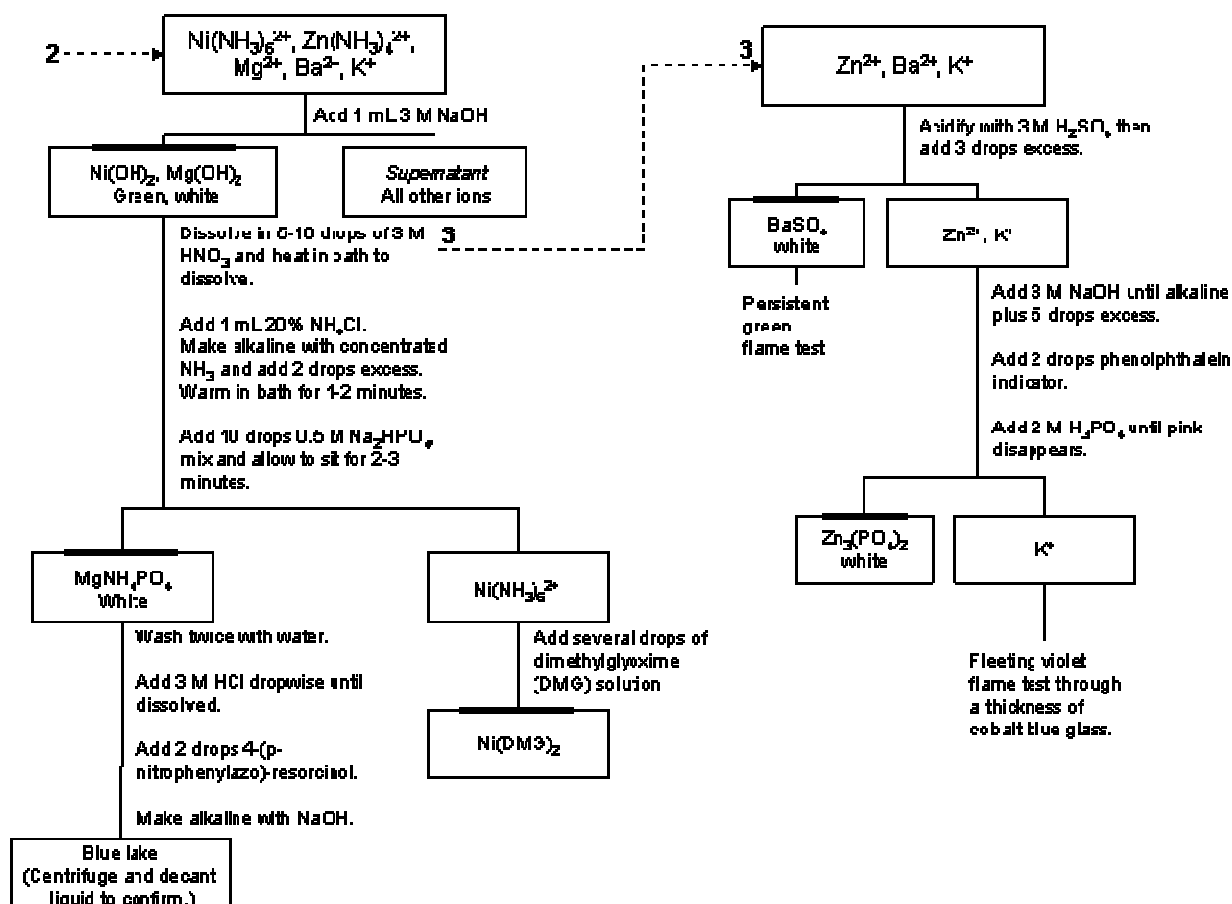
Flow chart for the qualitative analysis of cations from groups III and IV

Cation Separation Scheme

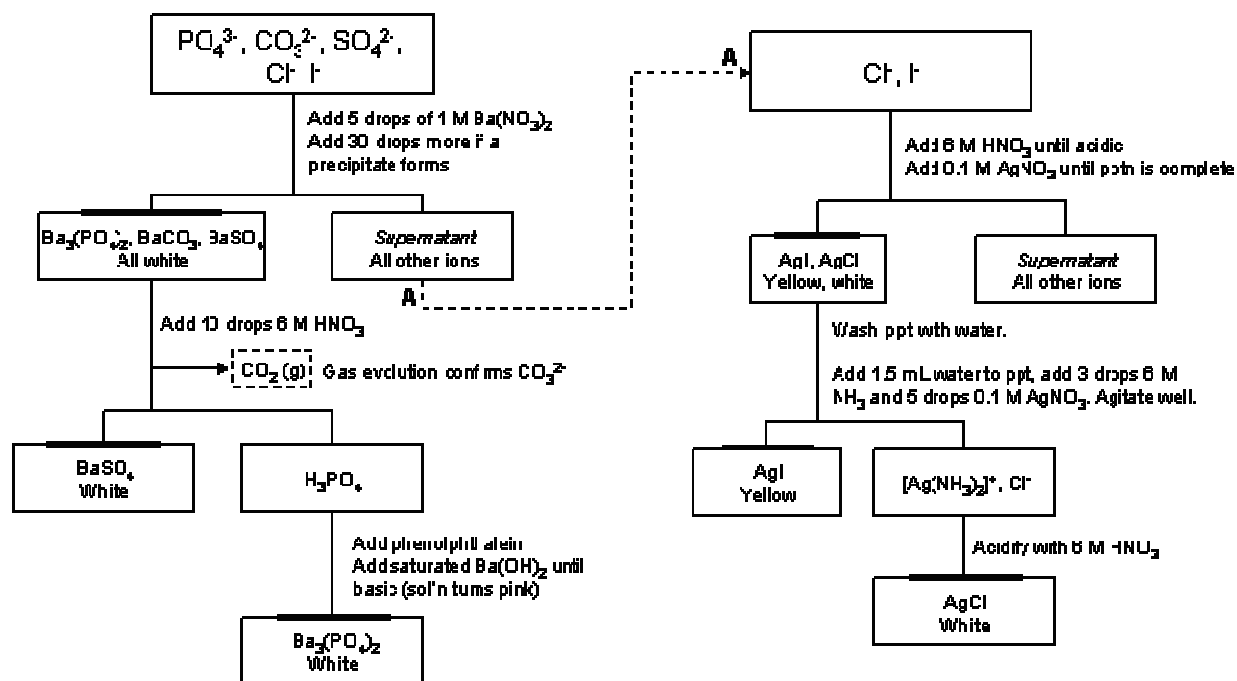


Used symbols:

- A.  Precipitate formed.
- B.  Solution containing ions.
- C.  Separation by centrifugation. After centrifuging, the supernatant should be transferred to a separate test tube and the precipitate should be washed with an appropriate solvent.
- D.  Divide solution and treat each part separately.



ANION SEPARATION SCHEME



Solubility of inorganic compounds at different temperature [g/100 g H₂O]

Sloučenina	0 °C	20 °C	60 °C	100 °C
AgF.2H ₂ O	-	432	771 (40 °C)	630 (108 °C)
AgNO ₂	0,16	0,34	1,36	-
AgNO ₃	122	222	525	952
Ag ₂ SO ₄	0,57	0,8	1,15	1,41
AlCl ₃ .6H ₂ O	123,4	131,9	142,8	147,2
Al(NO ₃) ₃ .9H ₂ O	194,5	297,5	976	§
Al ₂ (SO ₄) ₃ .18H ₂ O	86,2	108,2	262,7	1 108
Ba(C ₂ H ₃ O ₂) ₂ .3H ₂ O	81,7	101,2	106,3	108
BaCl ₂ .H ₂ O	39,2	44,6	59,2	76,8
Ba(HCO ₃) ₂	26,2	29,9	38,6	51,3
Ba(NO ₃) ₂	5	9,2	20,3	34,2
Ba(OH) ₂ .8H ₂ O	3,1	7,4	46,8	1268 (80 °C)
BaS ₂ O ₆ .2H ₂ O	7,8	15,7	19,86 (30 °C)	-
Br ₂	4,17	3,58	3,52 (50 °C)	
CaCl ₂ .2H ₂ O	97,6	130,1	325,9	435,2
CaCl ₂ .6H ₂ O	279,3	535,9	§	§
Ca(HCO ₃) ₂	16,2	16,6	17,5	18,4
CaCrO ₄	4,3	2,4	0,9	0,07
Ca(OH) ₂	0,18	0,16	0,12	0,08
Ca(NO ₃) ₂ .4H ₂ O	265,8	430,5	§	§
CaSO ₄ .2H ₂ O	0,22	0,26 (25 °C)	0,24 (65 °C)	0,2
CaSO ₄ .0,5H ₂ O	-	0,32	0,16	0,07
CdCl ₂ .H ₂ O	108,4	170,2	173,1	188,7
CdI ₂	79,8	86,2	97,4 (50 °C)	127,6
Cd(NO ₃) ₂ .4H ₂ O	255,7	399,1 (25 °C)	§	§
CoBr ₂ .6H ₂ O	253,4	337,7	-	-
CoCl ₂ .6H ₂ O	116,6	159,1	672,5	1 430

Sloučenina	0 °C	20 °C	60 °C	100 °C
$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	265,5	388,8	1 272	§
$\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$	58,6	93	215,4	464
$\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	-	265,7	-	-
CrO_3	164,9	-	182,1 (50 °C)	206,8
$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	110,6	123	152,9	192,4
$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	137,8	252	475,1	667,3 (80 °C)
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	24,3	36,6	80,8	205,3
$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$	159,7 (10 °C)	-	-	416,3
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	245,8	393,9	§	§
$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	205	350,6 (25 °C)	590,4 (40 °C)	-
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	32,7	62,1	149,0 (50 °C)	98,8 (90 °C)
$\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (šťavelová)	5,2	13,9	75	345 (90 °C)
$\text{H}_2\text{C}_4\text{H}_4\text{O}_4$ (vinná)	115	139	218	343
H_3BO_3	2,66	5,04	14,81	40,25
$\text{H}_3\text{C}_6\text{H}_5\text{O}_7 \cdot \text{H}_2\text{O}$ (citronová)	-	133	-	-
HgBr_2	0,3	0,5	1,6	4,9
HgCl_2	3,6	6,54	16,2	61,3
I_2	-	0,03	0,08	-
$\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	5,6	11,4	57,5	229,1 (90 °C)
KBr	53,5	65,2	85,5	104
KBrO_3	3,1	6,9	22,7	50
KCl	27,6	34	45,5	57
KClO_3	3,3	7,4	24,5	56,7
KClO_4	0,75	1,8	9	21,8
$\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$	-	24,39 (25 °C)	-	50
$\text{KF} \cdot 2\text{H}_2\text{O}$	100,1	373,6	1 938	3481(80 °C)
$\text{K}(\text{HCOO})$ (mravenčan)	290	335	455	790
KHCO_3	22,4	33,2	60	-
KH_2PO_4	14,2	22,3	49,4	-
KHSO_4	36,3	51,4	67,3 (40 °C)	121,6
KI	127,5	144	176	208
KIO_3	4,73	8,13	18,5	32,2
KMnO_4	2,83	6,4	22,2	-
KNO_2	278,8	298,4	334,9 (40 °C)	412,8
KNO_3	13,3	31,6	110	246
$\text{KNaC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$ (vinan)	26	66 (25 °C)	-	-
KOH	97	112	140 (50 °C)	178

Sloučenina	0 °C	20 °C	60 °C	100 °C
KSCN	177	217,5	-	-
K ₂ CO ₃	105,5	110,	126,8	155,7
K ₂ C ₂ O ₄ ·H ₂ O (šťavelan)	28,9	40,2	65,1	97,2
K ₂ CrO ₄	58,2	61,7	68,6	75,6
K ₂ Cr ₂ O ₇	5	12	43	80
K ₂ Cu(SO ₄) ₂ ·6H ₂ O	4,84	9,09	15,73 (40 °C)	-
K ₂ HPO ₄	-	-	262,8	279,8
K ₂ SO ₄	7,35	11,11	18,17	24,1
K ₂ Fe(SO ₄) ₂ ·6H ₂ O	28	48,1	94,5	-
K ₃ Fe(CN) ₆	31	43	66	82,6 (104 °C)
K ₄ Fe(CN) ₆ ·3H ₂ O	16,6	33,7	59,6 (50 °C)	95,4
LiCl	67	78,5	103	127,5
Li ₂ CO ₃	1,54	1,33	1,01	0,72
MgCl ₂ ·6H ₂ O	281,3	305,1	425	909
MgSO ₄ ·7H ₂ O	93,5 (10 °C)	117,2	249,1	491
MnCl ₂ ·4H ₂ O	156,5	201,4	451,6	534
Mn(NO ₃) ₂ ·6H ₂ O	426,3	1 659	§	§
MnSO ₄ ·4H ₂ O	105,5	132,7	110,1	59,9
NH ₄ Cl	29,4	37,2	55,2	77,3
NH ₄ Cr(SO ₄) ₂ ·12H ₂ O	-	21,2 (25 °C)	32,8 (40 °C)	-
NH ₄ Fe(SO ₄) ₂ ·12H ₂ O	-	124,8 (25 °C)	-	-
NH ₄ NO ₃	118,3	192	421	871
NH ₄ VO ₃	-	0,5	2,6	7,6
(NH ₄) ₂ C ₄ H ₄ O ₆ ·2H ₂ O	45	63	87	-
(NH ₄) ₂ CrO ₄	-	40,4 (30 °C)	-	-
(NH ₄) ₂ Cr ₂ O ₇	18	34,9	87,4	149,5
(NH ₄) ₂ Cu(SO ₄) ₂ ·6H ₂ O	16,7	28,6	79	229,9
(NH ₄) ₂ Fe(SO ₄) ₂ ·6H ₂ O	18,1	25,4 (10 °C)	65,1 (50 °C)	89,4 (70 °C)
(NH ₄) ₂ Ni(SO ₄) ₂ ·6H ₂ O	1,3	9,1	25	29,7
Na(CH ₃ COO)·3H ₂ O (octan)	79,1	111,2	2 737	§
NaCl	35,7	36	37,3	39,8
NaClO ₃	79	101	155	230
NaClO ₄ ·H ₂ O	259	349,8 (25 °C)	557 (55 °C)	732
NaCN·2H ₂ O	106,1	172,6	268,3 (30 °C)	-
NaHCO ₃	6,9	9,6	16,4	-
NaH ₂ PO ₄ ·2H ₂ O	86,9	150	-	-
NaNO ₂	72,1	84,5	104,1 (50 °C)	163,2

Sloučenina	0 °C	20 °C	60 °C	100 °C
NaNO ₃	73	88	124	180
NaOH	42	109	174	347
Na ₂ B ₄ O ₇ ·10H ₂ O	2,4	5,2	47	187,7
Na ₂ CO ₃ ·10H ₂ O	21,4	91,4	593	542
Na ₂ C ₂ O ₄ (šřavelan)	2,7	3,4	5	6,5
Na ₂ C ₄ H ₄ O ₆ ·2H ₂ O (vinan)	29 (6 °C)	-	66 (43 °C)	-
Na ₂ CrO ₄ ·10H ₂ O	103,4	13 807	§	§
Na ₂ Cr ₂ O ₇ ·2H ₂ O	238,9	267,6	638 (70 °C)	1171
Na ₂ HPO ₄ ·12H ₂ O	4,24	22,36	-	-
Na ₂ S·9H ₂ O	69,6 (10 °C)	94,9	1 162	§
Na ₂ SO ₃ ·7H ₂ O	32,2	73,6	80,9	-
Na ₂ SO ₄ ·10H ₂ O	12,1	58,3	241,4	209
Na ₂ S ₂ O ₃ ·5H ₂ O	117,5	182,7	8001 (50 °C)	§
NiCl ₂ ·6H ₂ O	179,6	253,4	479,5	596
Ni(NO ₃) ₂ ·6H ₂ O	239,4	356,2	7 423	§
NiSO ₄ ·7H ₂ O	63,4	118,2 (30 °C)	179,7	371,2
Pb(C ₂ H ₃ O ₂) ₂ ·3H ₂ O	23	51,6	135,25 (40 °C)	257,7 (50 °C)
PbBr ₂	0,46	0,85	2,36	4,8
PbCl ₂	0,76	0,99	1,98	3,3
PbI ₂	0,04	0,07	0,2	0,44
Pb(NO ₃) ₂	38,8	56,5	95	138,8
SbCl ₃	601,6	931,5	4 531	-
SnCl ₂ ·2H ₂ O	118,7	659 (15 °C)	-	-
ZnCl ₂	342	396	488	615
Zn(NO ₃) ₂ ·6H ₂ O	324,4	572,1	§	§
ZnSO ₄ ·7H ₂ O	110,9	168,4	341,8 (50 °C)	390,1

